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Schur's Triangularization Theorem and Sylvester's Theorem for Dual Complex Matrices

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ENDORSEMENT

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Abstract

Schur's Triangularization Theorem and Sylvester's Theorem for Dual Complex Matrices

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Dual complex numbers are numbers of the form $a_{\text{st}} + a_{\mathcal{I}}\varepsilon$ where a_{st} (called the *standard part*) and $a_{\mathcal{I}}$ (called the *infinitesimal part*) are complex numbers and ε is the dual unit satisfying the following properties: $\varepsilon \neq 0$ and $\varepsilon^2 = 0$. Dual complex matrices are matrices where all of its entries are dual complex numbers. One of the standard results in matrix analysis is Schur's triangularization theorem which states that a square matrix over $\mathbb{C}$ is unitarily similar to a triangular matrix whose diagonal entries are its eigenvalues. We show that Schur's triangularization theorem holds for $n \times n$ dual complex matrices whose standard part has n distinct eigenvalues. Some consequences of Schur's triangularization theorem are also examined, namely results about the trace and the determinant of dual complex matrices. Moreover, we also show that Sylvester's theorem on matrix equations can be extended to dual complex matrices, that is, $\widehat{A}\widehat{X} - \widehat{X}\widehat{B} = \widehat{C}$ has a unique solution $\widehat{X}$ if and only if the standard parts of $\widehat{A}$ and $\widehat{B}$ have no eigenvalues in common.

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Chapter 1

Dual Numbers over a Ring

Dual numbers have been around since the time of William K. Clifford [3] who discovered and studied them in 1873. Through time, some authors like Gungör & Tetik [4] and Qi & Luo [14] provided the definition of a dual complex number (i.e., numbers of the form $a_{\text{st}} + a_{\mathcal{I}}\varepsilon$ where a_{st} and $a_{\mathcal{I}}$ are complex numbers and ε is the dual unit) and a dual quaternion (i.e., numbers of the form $q_{\text{st}} + q_{\mathcal{I}}\varepsilon$ where q_{st} and $q_{\mathcal{I}}$ are quaternions), respectively. In this paper, we consider dual numbers over an arbitrary ring R. The following definition is similar to the definition of a dual number over $\mathbb{R}$ in [5] and [16].

Definition 1.1. Let R be a ring and let $a_{\text{st}}, a_{\mathcal{I}} \in R$. The numbers of the form $\widehat{a} = a_{\text{st}} + a_{\mathcal{I}}\varepsilon = a_{\text{st}} + \varepsilon a_{\mathcal{I}}$ are called *dual numbers* over R, where ε is called the *dual unit* satisfying the following properties: $\varepsilon \neq 0$ and $\varepsilon^2 = 0$. We call a_{st} and $a_{\mathcal{I}}$ the *standard part* and *infinitesimal part* of $\widehat{a}$, respectively. Moreover, we say that $\widehat{a}$ is *appreciable* if a_{st} is nonzero; otherwise, we say that $\widehat{a}$ is *infinitesimal*. We denote $\mathbb{D}R$ as the set of dual numbers over the ring R.

Remark 1.2.

1. The definition above requires $a\varepsilon = \varepsilon a$ for all $a \in R$.
2. Regarding the notation, we put the hat symbol $\widehat{\ }$ over a lowercase letter (not in boldface) to indicate a dual number over the ring R (e.g., $\widehat{a}$).
3. We assume that $a \in R$ can be written as $a = a + 0\varepsilon$. This means that $R \subseteq \mathbb{D}R$.
4. Two dual numbers over R are equal if and only if their standard parts and infinitesimal parts are equal, i.e., $\widehat{a} = \widehat{b}$ if and only if $a_{\text{st}} = b_{\text{st}}$ and $a_{\mathcal{I}} = b_{\mathcal{I}}$ for all $\widehat{a}, \widehat{b} \in \mathbb{D}R$.
5. In this thesis, a ring will be denoted by R and a field will be denoted by $\mathbb{F}$.

The following definition of addition and multiplication of dual numbers over R is similar to the definition in [10].

Definition 1.3. Let R be a ring and let $\widehat{a}, \widehat{b} \in \mathbb{D}R$. We define the addition and multiplication of dual numbers over the ring R as

$$\widehat{a} + \widehat{b} = \underbrace{(a_{\text{st}} + b_{\text{st}})}_{\in R} + \underbrace{(a_{\mathcal{I}} + b_{\mathcal{I}})}_{\in R}\varepsilon \text{ and}$$

$$\widehat{a}\widehat{b} = \underbrace{(a_{\text{st}}b_{\text{st}})}_{\in R} + \underbrace{(a_{\text{st}}b_{\mathcal{I}} + a_{\mathcal{I}}b_{\text{st}})}_{\in R}\varepsilon,$$

respectively.

Remark 1.4. Let $n \in \mathbb{N}$. By the above definition, we have the following consequences:

1. If R has no zero divisors, then the product of n appreciable dual numbers over R is always appreciable;
2. The product of n dual numbers over R with at least one infinitesimal dual number over R is always infinitesimal; and
3. The product of $n + 1$ infinitesimal dual numbers over R is always equal to zero.

We can see an example of these results from Proposition 4.4.

Furthermore, defining the addition and multiplication of dual numbers over the ring R is important to describe and examine their algebraic structure [9].

Proposition 1.5. *Let R be a ring. Then $\mathbb{D}R$ together with the addition defined in Definition 1.3 is an additive abelian group.*

Proof. Note that $\mathbb{D}R$ is nonempty since R is non-empty and $R \subseteq \mathbb{D}R$ (by Remark 1.2.3). Now, let $\widehat{a}, \widehat{b}, \widehat{c} \in \mathbb{D}R$. As emphasized in the Definition 1.3, $\widehat{a} + \widehat{b} \in \mathbb{D}R$. Thus $\mathbb{D}R$ is closed under addition of dual numbers over R. Furthermore, note that $\mathbb{D}R$ satisfies the following properties:

(G1) $\mathbb{D}R$ is associative under addition. Indeed, since R is associative under addition,

$$\begin{aligned}
\widehat{a} + (\widehat{b} + \widehat{c}) &= a_{\text{st}} + a_{\mathcal{I}}\varepsilon + ((b_{\text{st}} + b_{\mathcal{I}}\varepsilon) + (c_{\text{st}} + c_{\mathcal{I}}\varepsilon)) \\
&= a_{\text{st}} + a_{\mathcal{I}}\varepsilon + ((b_{\text{st}} + c_{\text{st}}) + (b_{\mathcal{I}} + c_{\mathcal{I}})\varepsilon) \\
&= a_{\text{st}} + (b_{\text{st}} + c_{\text{st}}) + (a_{\mathcal{I}} + (b_{\mathcal{I}} + c_{\mathcal{I}}))\varepsilon \\
&= (a_{\text{st}} + b_{\text{st}}) + c_{\text{st}} + ((a_{\mathcal{I}} + b_{\mathcal{I}}) + c_{\mathcal{I}})\varepsilon \\
&= ((a_{\text{st}} + a_{\mathcal{I}}\varepsilon) + (b_{\text{st}} + b_{\mathcal{I}}\varepsilon)) + c_{\text{st}} + c_{\mathcal{I}}\varepsilon \\
&= (\widehat{a} + \widehat{b}) + \widehat{c}.
\end{aligned}$$

(G2) $\mathbb{D}R$ has an additive identity. Indeed, by Remark 1.2.3, $0 = 0 + 0\varepsilon$. Hence

$$\begin{aligned} 0 + \widehat{a} &= (0 + 0\varepsilon) + (a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon) \\ &= (0 + a_{\mathrm{st}}) + (0 + a_{\mathcal{I}})\varepsilon \\ &= \widehat{a} \\ &= (a_{\mathrm{st}} + 0) + (a_{\mathcal{I}} + 0)\varepsilon \\ &= (a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon) + (0 + 0\varepsilon) \\ &= \widehat{a} + 0. \end{aligned}$$

(G3) Every element in $\mathbb{D}R$ has an additive inverse. Indeed, for every $\widehat{a} \in \mathbb{D}R$, define $-\widehat{a} = -a_{\mathrm{st}} - a_{\mathcal{I}}\varepsilon = -a_{\mathrm{st}} + (-a_{\mathcal{I}})\varepsilon$. Then

$$\begin{aligned} \widehat{a} + (-\widehat{a}) &= (a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon) + (-a_{\mathrm{st}} - a_{\mathcal{I}}\varepsilon) \\ &= (a_{\mathrm{st}} - a_{\mathrm{st}}) + (a_{\mathcal{I}} - a_{\mathcal{I}})\varepsilon \\ &= 0 \\ &= (-a_{\mathrm{st}} + a_{\mathrm{st}}) + (-a_{\mathcal{I}} + a_{\mathcal{I}})\varepsilon \\ &= (-a_{\mathrm{st}} - a_{\mathcal{I}}\varepsilon) + (a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon) \\ &= -\widehat{a} + \widehat{a}. \end{aligned}$$

(G4) $\mathbb{D}R$ is commutative under addition. Indeed, we have

$$\begin{aligned} \widehat{a} + \widehat{b} &= (a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon) + (b_{\mathrm{st}} + b_{\mathcal{I}}\varepsilon) \\ &= (a_{\mathrm{st}} + b_{\mathrm{st}}) + (a_{\mathcal{I}} + b_{\mathcal{I}})\varepsilon \\ &= (b_{\mathrm{st}} + a_{\mathrm{st}}) + (b_{\mathcal{I}} + a_{\mathcal{I}})\varepsilon \\ &= (b_{\mathrm{st}} + b_{\mathcal{I}}\varepsilon) + (a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon) \\ &= \widehat{b} + \widehat{a}. \end{aligned}$$

□

Proposition 1.6. *Let R be a ring. Then $\mathbb{D}R$ together with the addition and multiplication defined in Definition 1.3 forms a ring. Moreover, if R is commutative or R has unity, then $\mathbb{D}R$ is commutative or $\mathbb{D}R$ has unity, respectively.*

Proof. By Proposition 1.5, $\mathbb{D}R$ is an additive abelian group. Let $\widehat{a}, \widehat{b}, \widehat{c} \in \mathbb{D}R$.

Note that multiplication of dual numbers over R is associative since

$$\begin{aligned}
\widehat{a}(\widehat{b}\widehat{c}) &= (a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon)\,((b_{\mathrm{st}} + b_{\mathcal{I}}\varepsilon)(c_{\mathrm{st}} + c_{\mathcal{I}}\varepsilon)) \\
&= (a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon)((b_{\mathrm{st}}c_{\mathrm{st}}) + (b_{\mathrm{st}}c_{\mathcal{I}} + b_{\mathcal{I}}c_{\mathrm{st}})\varepsilon) \\
&= a_{\mathrm{st}}(b_{\mathrm{st}}c_{\mathrm{st}}) + (a_{\mathrm{st}}(b_{\mathrm{st}}c_{\mathcal{I}} + b_{\mathcal{I}}c_{\mathrm{st}}) + a_{\mathcal{I}}(b_{\mathrm{st}}c_{\mathrm{st}}))\varepsilon \\
&= a_{\mathrm{st}}(b_{\mathrm{st}}c_{\mathrm{st}}) + (a_{\mathrm{st}}(b_{\mathrm{st}}c_{\mathcal{I}}) + a_{\mathrm{st}}(b_{\mathcal{I}}c_{\mathrm{st}}) + a_{\mathcal{I}}(b_{\mathrm{st}}c_{\mathrm{st}}))\varepsilon \\
&= (a_{\mathrm{st}}b_{\mathrm{st}})c_{\mathrm{st}} + ((a_{\mathrm{st}}b_{\mathrm{st}})c_{\mathcal{I}} + (a_{\mathrm{st}}b_{\mathcal{I}})c_{\mathrm{st}} + (a_{\mathcal{I}}b_{\mathrm{st}})c_{\mathrm{st}})\varepsilon \\
&= (a_{\mathrm{st}}b_{\mathrm{st}})c_{\mathrm{st}} + ((a_{\mathrm{st}}b_{\mathrm{st}})c_{\mathcal{I}} + (a_{\mathrm{st}}b_{\mathcal{I}} + a_{\mathcal{I}}b_{\mathrm{st}})c_{\mathrm{st}})\varepsilon \\
&= (a_{\mathrm{st}}b_{\mathrm{st}} + (a_{\mathrm{st}}b_{\mathcal{I}} + a_{\mathcal{I}}b_{\mathrm{st}})\varepsilon)(c_{\mathrm{st}} + c_{\mathcal{I}}\varepsilon) \\
&= ((a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon)(b_{\mathrm{st}} + b_{\mathcal{I}}\varepsilon))(c_{\mathrm{st}} + c_{\mathcal{I}}\varepsilon) \\
&= (\widehat{a}\widehat{b})\widehat{c}.
\end{aligned}$$

Also,

$$\begin{aligned}
\widehat{a}(\widehat{b} + \widehat{c}) &= (a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon)((b_{\mathrm{st}} + b_{\mathcal{I}}\varepsilon) + (c_{\mathrm{st}} + c_{\mathcal{I}}\varepsilon)) \\
&= (a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon)(b_{\mathrm{st}} + c_{\mathrm{st}} + (b_{\mathcal{I}} + c_{\mathcal{I}})\varepsilon) \\
&= a_{\mathrm{st}}(b_{\mathrm{st}} + c_{\mathrm{st}}) + (a_{\mathrm{st}}(b_{\mathcal{I}} + c_{\mathcal{I}}) + a_{\mathcal{I}}(b_{\mathrm{st}} + c_{\mathrm{st}}))\varepsilon \\
&= a_{\mathrm{st}}b_{\mathrm{st}} + a_{\mathrm{st}}c_{\mathrm{st}} + (a_{\mathrm{st}}b_{\mathcal{I}} + a_{\mathrm{st}}c_{\mathcal{I}} + a_{\mathcal{I}}b_{\mathrm{st}} + a_{\mathcal{I}}c_{\mathrm{st}})\varepsilon \\
&= a_{\mathrm{st}}b_{\mathrm{st}} + a_{\mathrm{st}}c_{\mathrm{st}} + (a_{\mathrm{st}}b_{\mathcal{I}} + a_{\mathcal{I}}b_{\mathrm{st}} + a_{\mathrm{st}}c_{\mathcal{I}} + a_{\mathcal{I}}c_{\mathrm{st}})\varepsilon \\
&= a_{\mathrm{st}}b_{\mathrm{st}} + (a_{\mathrm{st}}b_{\mathcal{I}} + a_{\mathcal{I}}b_{\mathrm{st}})\varepsilon + a_{\mathrm{st}}c_{\mathrm{st}} + (a_{\mathrm{st}}c_{\mathcal{I}} + a_{\mathcal{I}}c_{\mathrm{st}})\varepsilon \\
&= (a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon)(b_{\mathrm{st}} + b_{\mathcal{I}}\varepsilon) + (a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon)(c_{\mathrm{st}} + c_{\mathcal{I}}\varepsilon) \\
&= \widehat{a}\widehat{b} + \widehat{a}\widehat{c}.
\end{aligned}$$

Similarly, we can show that $(\widehat{a} + \widehat{b})\widehat{c} = \widehat{a}\widehat{c} + \widehat{b}\widehat{c}$. Thus, the operations defined in Definition 1.3 follow the left and right distributive laws.
If R is commutative, then

$$\begin{aligned}
\widehat{a}\widehat{b} &= (a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon)(b_{\mathrm{st}} + b_{\mathcal{I}}\varepsilon) \\
&= a_{\mathrm{st}}b_{\mathrm{st}} + (a_{\mathrm{st}}b_{\mathcal{I}} + a_{\mathcal{I}}b_{\mathrm{st}})\varepsilon \\
&= b_{\mathrm{st}}a_{\mathrm{st}} + (b_{\mathrm{st}}a_{\mathcal{I}} + b_{\mathcal{I}}a_{\mathrm{st}})\varepsilon \\
&= (b_{\mathrm{st}} + b_{\mathcal{I}}\varepsilon)(a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon) \\
&= \widehat{b}\widehat{a}.
\end{aligned}$$

Lastly, if R has unity 1, then for all $\widehat{a} \in \mathbb{D}R$, we have

$$\begin{aligned} 1\widehat{a} &= (1 + 0\varepsilon)(a_{\text{st}} + a_{\mathcal{I}}\varepsilon) \\ &= 1a_{\text{st}} + (1a_{\mathcal{I}} + 0a_{\text{st}})\varepsilon \\ &= (a_{\text{st}} + a_{\mathcal{I}}\varepsilon)(1 + 0\varepsilon) \\ &= \widehat{a}1. \end{aligned}$$

□

Proposition 1.7. *Let R be a ring with unity 1 and let $\widehat{a} \in \mathbb{D}R$. Then*

1. *$\widehat{a}$ is a unit if and only if a_{st} is a unit; and*
2. *$\widehat{a}$ has a unique multiplicative inverse.*

Proof. ($\rightarrow$) Suppose $\widehat{a}$ is a unit. Then there exists $\widehat{b} \in \mathbb{D}R$ such that

$$\begin{aligned} \widehat{a}\widehat{b} &= (a_{\text{st}} + a_{\mathcal{I}}\varepsilon)(b_{\text{st}} + b_{\mathcal{I}}\varepsilon) \\ &= a_{\text{st}}b_{\text{st}} + (a_{\text{st}}b_{\mathcal{I}} + a_{\mathcal{I}}b_{\text{st}})\varepsilon \\ &= 1 \\ &= b_{\text{st}}a_{\text{st}} + (b_{\mathcal{I}}a_{\text{st}} + b_{\text{st}}a_{\mathcal{I}})\varepsilon \\ &= (b_{\text{st}} + b_{\mathcal{I}}\varepsilon)(a_{\text{st}} + a_{\mathcal{I}}\varepsilon) \\ &= \widehat{b}\widehat{a}. \end{aligned}$$

Thus, $a_{\text{st}}b_{\text{st}} = 1 = b_{\text{st}}a_{\text{st}}$, and so a_{st} is a unit.
($\leftarrow$) Suppose a_{st} is a unit and consider $a_{\text{st}}^{-1} - a_{\text{st}}^{-1}a_{\mathcal{I}}a_{\text{st}}^{-1}\varepsilon$. We have

$$\begin{aligned} (a_{\text{st}} + a_{\mathcal{I}}\varepsilon)(a_{\text{st}}^{-1} - a_{\text{st}}^{-1}a_{\mathcal{I}}a_{\text{st}}^{-1}\varepsilon) &= a_{\text{st}}a_{\text{st}}^{-1} + (-a_{\text{st}}a_{\text{st}}^{-1}a_{\mathcal{I}}a_{\text{st}}^{-1} + a_{\mathcal{I}}a_{\text{st}}^{-1})\varepsilon \\ &= 1 + (-a_{\mathcal{I}}a_{\text{st}}^{-1} + a_{\mathcal{I}}a_{\text{st}}^{-1})\varepsilon \\ &= 1 \\ &= 1 + (a_{\text{st}}^{-1}a_{\mathcal{I}} - a_{\text{st}}^{-1}a_{\mathcal{I}})\varepsilon \\ &= a_{\text{st}}^{-1}a_{\text{st}} + (a_{\text{st}}^{-1}a_{\mathcal{I}} - a_{\text{st}}^{-1}a_{\mathcal{I}}a_{\text{st}}^{-1}a_{\text{st}})\varepsilon \\ &= (a_{\text{st}}^{-1} - a_{\text{st}}^{-1}a_{\mathcal{I}}a_{\text{st}}^{-1}\varepsilon)(a_{\text{st}} + a_{\mathcal{I}}\varepsilon). \end{aligned}$$

Hence, $\widehat{a}$ is a unit. Moreover, since $\mathbb{D}R$ is a ring and the set of units of a ring forms a multiplicative group, $\widehat{a}$ has a unique inverse. □

Remark 1.8. We denote the multiplicative inverse of $\widehat{a}$ as $\widehat{a}^{-1}$. We use $\frac{1}{\widehat{a}}$ if in addition, R is a commutative ring with unity 1.

Definition 1.9. [8] Let R and S be rings. A function $f : R \to S$ is a *homomorphism* of rings provided that for all $a, b \in R$:

$$f(a+b) = f(a) + f(b) \text{ and } f(ab) = f(a)f(b).$$

A *monomorphism* (resp. *epimorphism*, *isomorphism*) of rings is a homomorphism of rings which is an injective (resp. surjective, bijective) map.

Theorem 1.10 (First Isomorphism Theorem for Rings)**.** *[8] If $f : R \to S$ is a homomorphism of rings, then f induces an isomorphism $R/Ker\ f \cong Im\ f$.*

We now discuss a result on how we can construct dual numbers over a ring R from a certain algebraic structure (e.g., a polynomial ring $R[x]$) similarly with [11].

Theorem 1.11. *Let R be a ring. Then $R[x]/\langle x^2\rangle \cong \mathbb{D}R$ (as rings).*

Proof. Let $f(x) = \sum_{i=0}^{n} a_i x^i \in R[x]$ and $g(x) = \sum_{j=0}^{m} b_j x^j \in R[x]$, where $n = \deg f$ and $m = \deg g$ whenever $f(x)$ and $g(x)$ are nonconstant. If $f(x)$ or $g(x)$ is a constant polynomial, let $a_1 = 0$ and $n = 1$ or $b_1 = 0$ and $m = 1$, respectively. Define $\Phi : R[x] \to \mathbb{D}R$ such that $\Phi(f(x)) = a_0 + a_1\varepsilon$ for all $f(x) \in R[x]$. Note that Φ is well-defined. Indeed, $f(x) = g(x)$ implies $n = m$ and $a_i = b_i$ for all $i \in \{0, 1, 2, ..., n\}$. Hence, by equality of dual numbers over a ring R, we have

$$\Phi(f(x)) = a_0 + a_1\varepsilon = b_0 + b_1\varepsilon = \Phi(g(x)).$$

Now, we show that Φ is a ring homomorphism. Indeed, note that $R[x]$ is a polynomial ring over R and $\mathbb{D}R$ is a ring. Moreover, if we write

$$f(x)+g(x) = \begin{cases} (a_0+b_0) + (a_1+b_1)x + \sum_{k=2}^{n}(a_k+b_k)x^k + \sum_{l=n+1}^{m} b_l x^l & \text{if } m > n \\ (a_0+b_0) + (a_1+b_1)x + \sum_{k=2}^{n}(a_k+b_k)x^k & \text{if } m = n \\ (a_0+b_0) + (a_1+b_1)x + \sum_{k=2}^{m}(a_k+b_k)x^k + \sum_{l=m+1}^{n} a_l x^l & \text{if } m < n \end{cases}$$

then we have

$$\begin{aligned} \Phi(f(x)+g(x)) &= (a_0+b_0) + (a_1+b_1)\varepsilon \\ &= a_0 + a_1\varepsilon + b_0 + b_1\varepsilon \\ &= \Phi(f(x)) + \Phi(g(x)). \end{aligned}$$

Next, note that

$$\begin{aligned}
\Phi\left(f\left(x\right)g\left(x\right)\right) &= \Phi\left(\left(\sum_{i=0}^{n} a_i x^i\right)\left(\sum_{j=0}^{m} b_j x^j\right)\right) \\
&= \Phi\left(\sum_{k=0}^{m+n}\sum_{i+j=k} a_i b_j x^k\right) \\
&= a_0 b_0 + \left(a_0 b_1 + a_1 b_0\right)\varepsilon \\
&= \left(a_0 + a_1\varepsilon\right)\left(b_0 + b_1\varepsilon\right) \\
&= \Phi\left(\sum_{i=0}^{n} a_i x^i\right)\Phi\left(\sum_{j=0}^{m} b_j x^j\right) \\
&= \Phi\left(f\left(x\right)\right)\Phi\left(g\left(x\right)\right).
\end{aligned}$$

Now, suppose $f\left(x\right) \in \ker\left(\Phi\right)$. Then $\Phi\left(f\left(x\right)\right) = a_0 + a_1\varepsilon = 0$, i.e., $a_0 = a_1 = 0$, which implies that

$$f\left(x\right) = a_2 x^2 + ... + a_n x^n = x^2\left(a_2 + ... + a_n x^{n-2}\right) \in \langle x^2\rangle.$$

Moreover, suppose $f\left(x\right) \in \langle x^2\rangle$. Then, $f\left(x\right) = x^2 g\left(x\right)$ for some $g\left(x\right) \in R\left[x\right]$. This implies that

$$\Phi\left(f\left(x\right)\right) = \Phi\left(x^2\right)\Phi\left(g\left(x\right)\right) = 0 \cdot \left(g\left(x\right)\right) = 0,$$

and so $f\left(x\right) \in \ker\left(\Phi\right)$. Thus, we have shown that $\ker\left(\Phi\right) = \langle x^2\rangle$. Note that $\mathrm{Im}\left(\Phi\right) = \mathbb{D}R$. Indeed, it is clear that $\mathrm{Im}\left(\Phi\right) \subseteq \mathbb{D}R$. Since for all $\widehat{a} = a_{\mathrm{st}} + a_{\mathcal{I}}\varepsilon \in \mathbb{D}R$, $\Phi(a_{\mathrm{st}} + a_{\mathcal{I}}x) = \widehat{a}$, and so $\mathbb{D}R \subseteq \mathrm{Im}\left(\Phi\right)$. Thus, by first isomorphism theorem for rings, $R\left[x\right]/\langle x^2\rangle \cong \mathbb{D}R$. □

Chapter 2

Dual Matrices over a Ring

In this chapter, we establish some properties of matrices over $\mathbb{D}R$. The following definition is similar to the definition of dual matrices in [16].

Definition 2.1. Let R be a ring. The matrices of the form $\widehat{A} = A_{\text{st}} + \varepsilon A_{\mathcal{I}} = A_{\text{st}} + A_{\mathcal{I}}\varepsilon \in M_{m,n}(\mathbb{D}R)$ where $A_{\text{st}}, A_{\mathcal{I}} \in M_{m,n}(R)$ and ε is the dual unit are called $m \times n$ *dual matrices* over R. We call A_{st} and $A_{\mathcal{I}}$ the *standard part* and the *infinitesimal part* of $\widehat{A}$, respectively, and we denote $M_{m,n}(\mathbb{D}R)$ to be the set of $m \times n$ dual matrices over R. We say that $\widehat{A}$ is *appreciable* if A_{st} is nonzero; otherwise, we say that $\widehat{A}$ is *infinitesimal*. Now, let $\widehat{\mathbf{x}} \in \mathbb{D}R^n$. Then, we say that $\widehat{\mathbf{x}}$ is a *dual (column) vector* over R of length n.

Remark 2.2.

1. In particular, the definition requires $A\varepsilon = \varepsilon A$ for all $A \in M_{m,n}(R)$.
2. We use the hat symbol on capital letters to emphasize that a given matrix is a dual matrix over R (e.g., $\widehat{A}$). Furthermore, to emphasize a dual vector over R, we put a hat symbol over a **boldface** lowercase letter (e.g., $\widehat{\mathbf{x}}$.)
3. Two dual matrices over R are equal if and only if their standard parts and infinitesimal parts are equal, i.e., $\widehat{A} = \widehat{B}$ if and only if $A_{\text{st}} = B_{\text{st}}$ and $A_{\mathcal{I}} = B_{\mathcal{I}}$ for all $\widehat{A}, \widehat{B} \in M_{m,n}(\mathbb{D}R)$.

Definition 2.3. Let $\widehat{A}, \widehat{B} \in M_{m,n}(\mathbb{D}R)$ and $\widehat{C} \in M_{n,p}(\mathbb{D}R)$. We define addition and multiplication of dual matrices over R as:

$$\widehat{A} + \widehat{B} = (A_{\text{st}} + \varepsilon A_{\mathcal{I}}) + (B_{\text{st}} + \varepsilon B_{\mathcal{I}}) = \underbrace{A_{\text{st}} + B_{\text{st}}}_{\in M_{m,n}(R)} + \varepsilon \underbrace{(A_{\mathcal{I}} + B_{\mathcal{I}})}_{\in M_{m,n}(R)},$$

$$\widehat{A}\widehat{C} = (A_{\text{st}} + \varepsilon A_{\mathcal{I}})(C_{\text{st}} + \varepsilon C_{\mathcal{I}}) = \underbrace{A_{\text{st}}C_{\text{st}}}_{\in M_{n,p}(R)} + \varepsilon \underbrace{(A_{\text{st}}C_{\mathcal{I}} + A_{\mathcal{I}}C_{\text{st}})}_{\in M_{n,p}(R)},$$

respectively.

Remark 2.4.

1. As emphasized in the preceding definition, $M_{m,n}(\mathbb{D}R)$ is closed under the usual matrix addition. Moreover, $M_n(\mathbb{D}R)$ is closed under the usual matrix multiplication.

2. We can write any $A \in M_{m,n}(R)$ as $A = A + \varepsilon 0_{m,n}$, where $0_{m,n}$ is the zero matrix of $M_{m,n}(R)$. Furthermore, we can write any $\varepsilon A \in M_{m,n}(\mathbb{D}R)$ as $\varepsilon A = 0_{m,n} + \varepsilon A$.

3. From our definition above, $(\widehat{A} + \widehat{B})_{\text{st}} = A_{\text{st}} + B_{\text{st}}$ and $(\widehat{A}\widehat{B})_{\text{st}} = A_{\text{st}}B_{\text{st}}$.

Proposition 2.5. *Let R be a ring. Then $M_{m,n}(\mathbb{D}R)$ together with the addition defined in Definition 2.3 is an additive abelian group.*

Proof. The zero matrix $0_{m,n} \in M_{m,n}(\mathbb{D}R)$, so $M_{m,n}(\mathbb{D}R)$ is non-empty and by Remark 2.4.1, $+$ is a binary operation on $M_{m,n}(\mathbb{D}R)$. Let $\widehat{A}, \widehat{B}, \widehat{C} \in M_{m,n}(\mathbb{D}R)$. Note that

(G1) $+$ is associative:

$$\begin{aligned}
\widehat{A} + \left(\widehat{B} + \widehat{C}\right) &= A_{\text{st}} + \varepsilon A_{\mathcal{I}} + (B_{\text{st}} + \varepsilon B_{\mathcal{I}} + C_{\text{st}} + \varepsilon C_{\mathcal{I}}) \\
&= A_{\text{st}} + \varepsilon A_{\mathcal{I}} + (B_{\text{st}} + C_{\text{st}} + \varepsilon (B_{\mathcal{I}} + C_{\mathcal{I}})) \\
&= A_{\text{st}} + B_{\text{st}} + C_{\text{st}} + \varepsilon (A_{\mathcal{I}} + (B_{\mathcal{I}} + C_{\mathcal{I}})) \\
&= A_{\text{st}} + B_{\text{st}} + C_{\text{st}} + \varepsilon ((A_{\mathcal{I}} + B_{\mathcal{I}}) + C_{\mathcal{I}}) \\
&= A_{\text{st}} + B_{\text{st}} + \varepsilon (A_{\mathcal{I}} + B_{\mathcal{I}}) + C_{\text{st}} + \varepsilon C_{\mathcal{I}} \\
&= (A_{\text{st}} + \varepsilon A_{\mathcal{I}} + B_{\text{st}} + \varepsilon B_{\mathcal{I}}) + C_{\text{st}} + \varepsilon C_{\mathcal{I}} \\
&= \left(\widehat{A} + \widehat{B}\right) + \widehat{C};
\end{aligned}$$

(G2) $M_{m,n}(\mathbb{D}R)$ has an additive identity:

$$\begin{aligned}
\widehat{A} + 0_{m,n} &= A_{\text{st}} + \varepsilon A_{\mathcal{I}} + 0_{m,n} + \varepsilon 0_{m,n} \\
&= A_{\text{st}} + \varepsilon A_{\mathcal{I}} \\
&= \widehat{A} \\
&= 0_{m,n} + \varepsilon 0_{m,n} + A_{\text{st}} + \varepsilon A_{\mathcal{I}} \\
&= 0_{m,n} + \widehat{A};
\end{aligned}$$

(G3) Every $\widehat{A} \in M_{m,n}(\mathbb{D}R)$ has an additive inverse $-\widehat{A} = -A_{\text{st}} + \varepsilon(-A_{\mathcal{I}})$:

$$\begin{aligned}
\widehat{A} + \left(-\widehat{A}\right) &= A_{\text{st}} + \varepsilon A_{\mathcal{I}} + (-A_{\text{st}} + \varepsilon\,(-A_{\mathcal{I}})) \\
&= (A_{\text{st}} - A_{\text{st}}) + \varepsilon(A_{\mathcal{I}} - A_{\mathcal{I}}) \\
&= 0_{m,n} + \varepsilon 0_{m,n} \\
&= 0_{m,n} \\
&= (-A_{\text{st}} + A_{\text{st}}) + \varepsilon(-A_{\mathcal{I}} + A_{\mathcal{I}}) \\
&= (-A_{\text{st}} + \varepsilon\,(-A_{\mathcal{I}})) + A_{\text{st}} + \varepsilon A_{\mathcal{I}} \\
&= \left(-\widehat{A}\right) + \widehat{A};\ \text{and}
\end{aligned}$$

(G4) + is commutative:

$$\begin{aligned}
\widehat{A} + \widehat{B} &= A_{\text{st}} + \varepsilon A_{\mathcal{I}} + B_{\text{st}} + \varepsilon B_{\mathcal{I}} \\
&= A_{\text{st}} + B_{\text{st}} + \varepsilon\,(A_{\mathcal{I}} + B_{\mathcal{I}}) \\
&= B_{\text{st}} + A_{\text{st}} + \varepsilon\,(B_{\mathcal{I}} + A_{\mathcal{I}}) \\
&= B_{\text{st}} + \varepsilon B_{\mathcal{I}} + A_{\text{st}} + \varepsilon A_{\mathcal{I}} \\
&= \widehat{B} + \widehat{A}.
\end{aligned}$$

Hence, $M_{m,n}(\mathbb{D}R)$ forms a group under addition of dual matrices over R. □

Definition 2.6. [8] A (left) *R-module* is an additive abelian group A together with a function $R \times A \to A$ (the image of (r, a) being denoted by ra) such that for all $r, s \in R$ and $a, b \in A$:

(M1) $r(a + b) = ra + rb$;

(M2) $(r + s)a = ra + sa$;

(M3) $r(sa) = (rs)a$.

If R has a multiplicative identity 1 and

(M4) $1a = a$ for all $a \in A$,

then A is said to be a *unital* R-module.

Proposition 2.7. *Let R be a ring. Then $M_{m,n}(\mathbb{D}R)$ has a $\mathbb{D}R$-module structure, where for all $\widehat{r} = r_{st} + r_{\mathcal{I}}\varepsilon \in \mathbb{D}R$, $\widehat{A} = A_{st} + \varepsilon A_{\mathcal{I}} \in M_{m,n}(\mathbb{D}R)$, and $\widehat{B} = B_{st} + \varepsilon B_{\mathcal{I}} \in M_{m,n}(\mathbb{D}R)$,*

$$\widehat{r}\widehat{A} = r_{st}A_{st} + \varepsilon\,(r_{st}A_{\mathcal{I}} + r_{\mathcal{I}}A_{st})\,.$$

Moreover, if R has unity 1, then $M_{m,n}(\mathbb{D}R)$ is a unital $\mathbb{D}R-$module.

Proof. Clearly, $\widehat{r}\widehat{A} \in M_{m,n}(\mathbb{D}R)$. Also, $\mathbb{D}R$ is a ring and by Proposition 2.5, $M_{m,n}(\mathbb{D}R)$ is an additive abelian group. Thus, for all $\widehat{r}, \widehat{s} \in \mathbb{D}R$ and $\widehat{A}, \widehat{B} \in M_{m,n}(\mathbb{D}R)$, we have

(M1)

$$\begin{aligned}
\widehat{r}\left(\widehat{A}+\widehat{B}\right) &= (r_{\mathrm{st}}+r_{\mathcal{I}}\varepsilon)(A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}}+B_{\mathrm{st}}+\varepsilon B_{\mathcal{I}}) \\
&= (r_{\mathrm{st}}+r_{\mathcal{I}}\varepsilon)(A_{\mathrm{st}}+B_{\mathrm{st}}+\varepsilon(A_{\mathcal{I}}+B_{\mathcal{I}})) \\
&= r_{\mathrm{st}}(A_{\mathrm{st}}+B_{\mathrm{st}})+\varepsilon(r_{\mathrm{st}}(A_{\mathcal{I}}+B_{\mathcal{I}})+r_{\mathcal{I}}(A_{\mathrm{st}}+B_{\mathrm{st}})) \\
&= r_{\mathrm{st}}A_{\mathrm{st}}+r_{\mathrm{st}}B_{\mathrm{st}}+\varepsilon(r_{\mathrm{st}}A_{\mathcal{I}}+r_{\mathrm{st}}B_{\mathcal{I}}+r_{\mathcal{I}}A_{\mathrm{st}}+r_{\mathcal{I}}B_{\mathrm{st}}) \\
&= r_{\mathrm{st}}A_{\mathrm{st}}+r_{\mathrm{st}}B_{\mathrm{st}}+\varepsilon(r_{\mathrm{st}}A_{\mathcal{I}}+r_{\mathcal{I}}A_{\mathrm{st}}+r_{\mathrm{st}}B_{\mathcal{I}}+r_{\mathcal{I}}B_{\mathrm{st}}) \\
&= r_{\mathrm{st}}A_{\mathrm{st}}+\varepsilon(r_{\mathrm{st}}A_{\mathcal{I}}+r_{\mathcal{I}}A_{\mathrm{st}})+r_{\mathrm{st}}B_{\mathrm{st}}+\varepsilon(r_{\mathrm{st}}B_{\mathcal{I}}+r_{\mathcal{I}}B_{\mathrm{st}}) \\
&= (r_{\mathrm{st}}+r_{\mathcal{I}}\varepsilon)(A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}})+(r_{\mathrm{st}}+r_{\mathcal{I}}\varepsilon)(B_{\mathrm{st}}+\varepsilon B_{\mathcal{I}}) \\
&= \widehat{r}\widehat{A}+\widehat{r}\widehat{B};
\end{aligned}$$

(M2)

$$\begin{aligned}
(\widehat{r}+\widehat{s})\widehat{A} &= (r_{\mathrm{st}}+r_{\mathcal{I}}\varepsilon+s_{\mathrm{st}}+s_{\mathcal{I}}\varepsilon)(A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}}) \\
&= (r_{\mathrm{st}}+s_{\mathrm{st}}+(r_{\mathcal{I}}+s_{\mathcal{I}})\varepsilon)(A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}}) \\
&= (r_{\mathrm{st}}+s_{\mathrm{st}})A_{\mathrm{st}}+\varepsilon((r_{\mathrm{st}}+s_{\mathrm{st}})A_{\mathcal{I}}+(r_{\mathcal{I}}+s_{\mathcal{I}})A_{\mathrm{st}}) \\
&= r_{\mathrm{st}}A_{\mathrm{st}}+s_{\mathrm{st}}A_{\mathrm{st}}+\varepsilon((r_{\mathrm{st}}A_{\mathcal{I}}+s_{\mathrm{st}}A_{\mathcal{I}})+(r_{\mathcal{I}}A_{\mathrm{st}}+s_{\mathcal{I}}A_{\mathrm{st}})) \\
&= r_{\mathrm{st}}A_{\mathrm{st}}+s_{\mathrm{st}}A_{\mathrm{st}}+\varepsilon((r_{\mathrm{st}}A_{\mathcal{I}}+r_{\mathcal{I}}A_{\mathrm{st}})+(s_{\mathrm{st}}A_{\mathcal{I}}+s_{\mathcal{I}}A_{\mathrm{st}})) \\
&= r_{\mathrm{st}}A_{\mathrm{st}}+\varepsilon(r_{\mathrm{st}}A_{\mathcal{I}}+r_{\mathcal{I}}A_{\mathrm{st}})+s_{\mathrm{st}}A_{\mathrm{st}}+\varepsilon(s_{\mathrm{st}}A_{\mathcal{I}}+s_{\mathcal{I}}A_{\mathrm{st}}) \\
&= (r_{\mathrm{st}}+r_{\mathcal{I}}\varepsilon)(A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}})+(s_{\mathrm{st}}+s_{\mathcal{I}}\varepsilon)(A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}}) \\
&= \widehat{r}\widehat{A}+\widehat{s}\widehat{A};
\end{aligned}$$

(M3)

$$\begin{aligned}
\widehat{r}\left(\widehat{s}\widehat{A}\right) &= (r_{\mathrm{st}}+r_{\mathcal{I}}\varepsilon)((s_{\mathrm{st}}+s_{\mathcal{I}}\varepsilon)(A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}})) \\
&= (r_{\mathrm{st}}+r_{\mathcal{I}}\varepsilon)(s_{\mathrm{st}}A_{\mathrm{st}}+\varepsilon(s_{\mathrm{st}}A_{\mathcal{I}}+s_{\mathcal{I}}A_{\mathrm{st}})) \\
&= r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathrm{st}}+\varepsilon(r_{\mathrm{st}}(s_{\mathrm{st}}A_{\mathcal{I}}+s_{\mathcal{I}}A_{\mathrm{st}})+r_{\mathcal{I}}s_{\mathrm{st}}A_{\mathrm{st}}) \\
&= r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathrm{st}}+\varepsilon((r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathcal{I}}+r_{\mathrm{st}}s_{\mathcal{I}}A_{\mathrm{st}})+r_{\mathcal{I}}s_{\mathrm{st}}A_{\mathrm{st}}) \\
&= r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathrm{st}}+\varepsilon(r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathcal{I}}+(r_{\mathrm{st}}s_{\mathcal{I}}+r_{\mathcal{I}}s_{\mathrm{st}})A_{\mathrm{st}}) \\
&= (r_{\mathrm{st}}s_{\mathrm{st}}+\varepsilon(r_{\mathrm{st}}s_{\mathcal{I}}+r_{\mathcal{I}}s_{\mathrm{st}}))(A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}}) \\
&= ((r_{\mathrm{st}}+r_{\mathcal{I}}\varepsilon)(s_{\mathrm{st}}+s_{\mathcal{I}}\varepsilon))(A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}})
\end{aligned}$$

$$= (\widehat{rs})\,\widehat{A}.$$

Hence, $M_{m,n}(\mathbb{D}R)$ has a $\mathbb{D}R$-module structure.

It is easy to see that $M_{m,n}(\mathbb{D}R)$ has an R-module structure for some function $R \times M_{m,n}(\mathbb{D}R) \to M_{m,n}(\mathbb{D}R)$ since $R \subseteq \mathbb{D}R$. Moreover, if R has unity 1, observe that

$$1\widehat{A} = (1 + 0\varepsilon)\,(A_{\text{st}} + \varepsilon A_{\mathcal{I}}) = 1A_{\text{st}} + \varepsilon(1A_{\mathcal{I}} + 0A_{\text{st}}) = A_{\text{st}} + \varepsilon A_{\mathcal{I}} = \widehat{A}$$

for all $\widehat{A} \in M_{m,n}(\mathbb{D}R)$. Therefore, $M_{m,n}(\mathbb{D}R)$ is a unital $\mathbb{D}R-$module. □

Proposition 2.8. *Let R be a ring. Then $M_n(\mathbb{D}R)$ is a ring under the matrix addition and multiplication defined in Definition 2.3.*

Proof. The zero matrix $0_{n,n} \in M_n(\mathbb{D}R)$. Thus, $M_n(\mathbb{D}R)$ is non-empty. By Remark 2.4.1, $M_n(\mathbb{D}R)$ is closed under the matrix addition and multiplication defined in Definition 2.3. Moreover, $M_n(\mathbb{D}R)$ satisfies the 3 ring axioms:

(R1) $M_n(\mathbb{D}R)$ is an additive abelian group (by Proposition 2.5);

(R2) Multiplication in $M_n(\mathbb{D}R)$ is associative, i.e.,

$$\begin{aligned}
\left(\widehat{A}\widehat{B}\right)\widehat{C} &= \left((A_{\text{st}} + \varepsilon A_{\mathcal{I}})\,(B_{\text{st}} + \varepsilon B_{\mathcal{I}})\right)(C_{\text{st}} + \varepsilon C_{\mathcal{I}}) \\
&= \left(A_{\text{st}}B_{\text{st}} + \varepsilon\left(A_{\text{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\text{st}}\right)\right)(C_{\text{st}} + \varepsilon C_{\mathcal{I}}) \\
&= A_{\text{st}}B_{\text{st}}C_{\text{st}} + \varepsilon\left(A_{\text{st}}B_{\text{st}}C_{\mathcal{I}} + \left(A_{\text{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\text{st}}\right)C_{\text{st}}\right) \\
&= A_{\text{st}}B_{\text{st}}C_{\text{st}} + \varepsilon\left(A_{\text{st}}B_{\text{st}}C_{\mathcal{I}} + \left(A_{\text{st}}B_{\mathcal{I}}C_{\text{st}} + A_{\mathcal{I}}B_{\text{st}}C_{\text{st}}\right)\right) \\
&= A_{\text{st}}B_{\text{st}}C_{\text{st}} + \varepsilon\left(A_{\text{st}}\left(B_{\text{st}}C_{\mathcal{I}} + B_{\mathcal{I}}C_{\text{st}}\right) + A_{\mathcal{I}}B_{\text{st}}C_{\text{st}}\right) \\
&= (A_{\text{st}} + \varepsilon A_{\mathcal{I}})\left(B_{\text{st}}C_{\text{st}} + \varepsilon\left(B_{\text{st}}C_{\mathcal{I}} + B_{\mathcal{I}}C_{\text{st}}\right)\right) \\
&= (A_{\text{st}} + \varepsilon A_{\mathcal{I}})\left((B_{\text{st}} + \varepsilon B_{\mathcal{I}})\,(C_{\text{st}} + \varepsilon C_{\mathcal{I}})\right) \\
&= \widehat{A}\left(\widehat{B}\widehat{C}\right)
\end{aligned}$$

for all $\widehat{A}, \widehat{B}, \widehat{C} \in M_n(\mathbb{D}R)$;

(R3) $M_n(\mathbb{D}R)$ follows the left and right distributive laws, i.e.,

$$\begin{aligned}
\widehat{A}\left(\widehat{B}+\widehat{C}\right) &= (A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}})\,(B_{\mathrm{st}}+\varepsilon B_{\mathcal{I}}+C_{\mathrm{st}}+\varepsilon C_{\mathcal{I}}) \\
&= (A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}})\,(B_{\mathrm{st}}+C_{\mathrm{st}}+\varepsilon\,(B_{\mathcal{I}}+C_{\mathcal{I}})) \\
&= A_{\mathrm{st}}\,(B_{\mathrm{st}}+C_{\mathrm{st}})+\varepsilon\,(A_{\mathrm{st}}\,(B_{\mathcal{I}}+C_{\mathcal{I}})+A_{\mathcal{I}}\,(B_{\mathrm{st}}+C_{\mathrm{st}})) \\
&= A_{\mathrm{st}}B_{\mathrm{st}}+A_{\mathrm{st}}C_{\mathrm{st}}+\varepsilon\,(A_{\mathrm{st}}B_{\mathcal{I}}+A_{\mathrm{st}}C_{\mathcal{I}}+A_{\mathcal{I}}B_{\mathrm{st}}+A_{\mathcal{I}}C_{\mathrm{st}}) \\
&= A_{\mathrm{st}}B_{\mathrm{st}}+A_{\mathrm{st}}C_{\mathrm{st}}+\varepsilon\,(A_{\mathrm{st}}B_{\mathcal{I}}+A_{\mathcal{I}}B_{\mathrm{st}}+A_{\mathrm{st}}C_{\mathcal{I}}+A_{\mathcal{I}}C_{\mathrm{st}}) \\
&= A_{\mathrm{st}}B_{\mathrm{st}}+\varepsilon\,(A_{\mathrm{st}}B_{\mathcal{I}}+A_{\mathcal{I}}B_{\mathrm{st}})+A_{\mathrm{st}}C_{\mathrm{st}}+\varepsilon\,(A_{\mathrm{st}}C_{\mathcal{I}}+A_{\mathcal{I}}C_{\mathrm{st}}) \\
&= (A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}})\,(B_{\mathrm{st}}+\varepsilon B_{\mathcal{I}})+(A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}})\,(C_{\mathrm{st}}+\varepsilon C_{\mathcal{I}}) \\
&= \widehat{A}\widehat{B}+\widehat{A}\widehat{C}
\end{aligned}$$

and

$$\begin{aligned}
\left(\widehat{A}+\widehat{B}\right)\widehat{C} &= (A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}}+B_{\mathrm{st}}+\varepsilon B_{\mathcal{I}})\,(C_{\mathrm{st}}+\varepsilon C_{\mathcal{I}}) \\
&= (A_{\mathrm{st}}+B_{\mathrm{st}}+\varepsilon\,(A_{\mathcal{I}}+B_{\mathcal{I}}))\,(C_{\mathrm{st}}+\varepsilon C_{\mathcal{I}}) \\
&= (A_{\mathrm{st}}+B_{\mathrm{st}})\,C_{\mathrm{st}}+\varepsilon\,((A_{\mathrm{st}}+B_{\mathrm{st}})\,C_{\mathcal{I}}+(A_{\mathcal{I}}+B_{\mathcal{I}})\,C_{\mathrm{st}}) \\
&= A_{\mathrm{st}}C_{\mathrm{st}}+B_{\mathrm{st}}C_{\mathrm{st}}+\varepsilon\,(A_{\mathrm{st}}C_{\mathcal{I}}+B_{\mathrm{st}}C_{\mathcal{I}}+A_{\mathcal{I}}C_{\mathrm{st}}+B_{\mathcal{I}}C_{\mathrm{st}}) \\
&= A_{\mathrm{st}}C_{\mathrm{st}}+B_{\mathrm{st}}C_{\mathrm{st}}+\varepsilon\,(A_{\mathrm{st}}C_{\mathcal{I}}+A_{\mathcal{I}}C_{\mathrm{st}}+B_{\mathrm{st}}C_{\mathcal{I}}+B_{\mathcal{I}}C_{\mathrm{st}}) \\
&= A_{\mathrm{st}}C_{\mathrm{st}}+\varepsilon\,(A_{\mathrm{st}}C_{\mathcal{I}}+A_{\mathcal{I}}C_{\mathrm{st}})+B_{\mathrm{st}}C_{\mathrm{st}}+\varepsilon\,(B_{\mathrm{st}}C_{\mathcal{I}}+B_{\mathcal{I}}C_{\mathrm{st}}) \\
&= (A_{\mathrm{st}}+\varepsilon A_{\mathcal{I}})\,(C_{\mathrm{st}}+\varepsilon C_{\mathcal{I}})+(B_{\mathrm{st}}+\varepsilon B_{\mathcal{I}})\,(C_{\mathrm{st}}+\varepsilon C_{\mathcal{I}}) \\
&= \widehat{A}\widehat{C}+\widehat{B}\widehat{C}
\end{aligned}$$

for all $\widehat{A},\widehat{B},\widehat{C}\in M_{m,n}(\mathbb{D}R)$.

Hence, $M_n(\mathbb{D}R)$ is a ring. □

Definition 2.9. [8] Let K be a commutative ring with unity. A *K-algebra* (or *algebra* over K) A is a ring A such that:

(A1) $(A,+)$ is a unital (left) K-module; and

(A2) $k(ab)=(ka)b=a(kb)$ for all $k\in K$ and $a,b\in A$.

If the K-algebra A is a division ring, we say that A is a *division algebra*.

Proposition 2.10. *Let R be a commutative ring with unity. Then $M_n(\mathbb{D}R)$ has a $\mathbb{D}R$-algebra structure.*

Proof. Since R is a commutative ring with unity, $\mathbb{D}R$ is a commutative ring with unity and by Proposition 2.8, $M_n(\mathbb{D}R)$ is a ring. Thus, we only need to show that $M_n(\mathbb{D}R)$ satisfies the properties (A1) and (A2) of Definition 2.9. Indeed,

(A1) $(M_n(\mathbb{D}R), +)$ is a unital $\mathbb{D}R$-module (by Proposition 2.7).

(A2) Since R is commutative, $kAB = (kA)B = A(kB)$ for all $A, B \in M_n(\mathbb{R})$ and for all $k \in R$. Hence, for all $\widehat{k} \in \mathbb{D}R$ and $\widehat{A}, \widehat{B} \in M_n(\mathbb{D}R)$, we have

$$\begin{aligned}
\widehat{k}\left(\widehat{A}\widehat{B}\right) &= (k_{\mathrm{st}} + k_{\mathcal{I}}\varepsilon)\left((A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}})(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}})\right) \\
&= (k_{\mathrm{st}} + k_{\mathcal{I}}\varepsilon)\left(A_{\mathrm{st}}B_{\mathrm{st}} + \varepsilon\left(A_{\mathrm{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\mathrm{st}}\right)\right) \\
&= k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathrm{st}} + \varepsilon\left(k_{\mathrm{st}}\left(A_{\mathrm{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\mathrm{st}}\right) + k_{\mathcal{I}}A_{\mathrm{st}}B_{\mathrm{st}}\right) \\
&= k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathrm{st}} + \varepsilon\left(k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathcal{I}} + k_{\mathrm{st}}A_{\mathcal{I}}B_{\mathrm{st}} + k_{\mathcal{I}}A_{\mathrm{st}}B_{\mathrm{st}}\right) \\
&= k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathrm{st}} + \varepsilon\left(k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathcal{I}} + \left(k_{\mathrm{st}}A_{\mathcal{I}} + k_{\mathcal{I}}A_{\mathrm{st}}\right)B_{\mathrm{st}}\right) \\
&= \left(k_{\mathrm{st}}A_{\mathrm{st}} + \varepsilon\left(k_{\mathrm{st}}A_{\mathcal{I}} + k_{\mathcal{I}}A_{\mathrm{st}}\right)\right)\left(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}\right) \\
&= \left(\left(k_{\mathrm{st}} + k_{\mathcal{I}}\varepsilon\right)\left(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}}\right)\right)\left(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}\right) \\
&= \left(\widehat{k}\widehat{A}\right)\widehat{B} \\
&= k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathrm{st}} + \varepsilon\left(k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathcal{I}} + k_{\mathcal{I}}A_{\mathrm{st}}B_{\mathrm{st}} + k_{\mathrm{st}}A_{\mathcal{I}}B_{\mathrm{st}}\right) \\
&= \left(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}}\right)\left(k_{\mathrm{st}}B_{\mathrm{st}} + \varepsilon\left(k_{\mathrm{st}}B_{\mathcal{I}} + k_{\mathcal{I}}B_{\mathrm{st}}\right)\right) \\
&= \left(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}}\right)\left(\left(k_{\mathrm{st}} + k_{\mathcal{I}}\varepsilon\right)\left(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}\right)\right) \\
&= \widehat{A}\left(\widehat{k}\widehat{B}\right).
\end{aligned}$$

Therefore, $M_n(\mathbb{D}R)$ has a $\mathbb{D}R$-algebra structure. □

Proposition 2.11. *Let R be a ring. Define*

$$\mathscr{D}_{m,n}(R) := \left\{ \mathcal{A} \in M_{2m,2n}(R) \;\middle|\; \mathcal{A} = \begin{pmatrix} A_{st} & A_{\mathcal{I}} \\ 0 & A_{st} \end{pmatrix},\ \textit{where } A_{st}, A_{\mathcal{I}} \in M_{m,n}(R) \right\}.$$

Then, $\mathscr{D}_{m,n}(R)$ is an abelian group under the usual block matrix addition.

Proof. Note that $\mathscr{D}_{m,n}(R)$ is non-empty since the zero matrix $0_{2m,2n} \in \mathscr{D}_{m,n}(R)$. Let $\mathcal{A} = \begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} \in \mathscr{D}_{m,n}(R)$ and $\mathcal{B} = \begin{pmatrix} B_{\mathrm{st}} & B_{\mathcal{I}} \\ 0 & B_{\mathrm{st}} \end{pmatrix} \in \mathscr{D}_{m,n}(R)$. Since $A_{\mathrm{st}} - B_{\mathrm{st}}, A_{\mathcal{I}} - B_{\mathcal{I}} \in M_{m,n}(R)$, we have

$$\mathcal{A} - \mathcal{B} = \begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} - \begin{pmatrix} B_{\mathrm{st}} & B_{\mathcal{I}} \\ 0 & B_{\mathrm{st}} \end{pmatrix} = \begin{pmatrix} A_{\mathrm{st}} - B_{\mathrm{st}} & A_{\mathcal{I}} - B_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} - B_{\mathrm{st}} \end{pmatrix} \in \mathscr{D}_{m,n}(R).$$

Thus, by subgroup test, $\mathscr{D}_{m,n}(R)$ is a subgroup of the additive abelian group $M_{2m,2n}(R)$. Hence, $\mathscr{D}_{m,n}(R)$ is an abelian group under the usual block matrix addition. □

Proposition 2.12. *Let R be a ring. Then $\mathscr{D}_{m,n}(R)$ has a $\mathbb{D}R$-module structure, where*

$$\widehat{r} * \mathcal{A} := \begin{pmatrix} r_{st}A_{st} & r_{\mathcal{I}}A_{st} + r_{st}A_{\mathcal{I}} \\ 0 & r_{st}A_{st} \end{pmatrix}$$

for all $\widehat{r} \in \mathbb{D}R$ and $\mathcal{A} = \begin{pmatrix} A_{st} & A_{\mathcal{I}} \\ 0 & A_{st} \end{pmatrix} \in \mathscr{D}_{m,n}(R)$.

Proof. By Proposition 1.6, $\mathbb{D}R$ is a ring. Moreover, by Proposition 2.11, $\mathscr{D}_{m,n}(R)$ is an additive abelian group. Let $\widehat{r}, \widehat{s} \in \mathbb{D}R$ and $\mathcal{A} = \begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix}, \mathcal{B} = \begin{pmatrix} B_{\mathrm{st}} & B_{\mathcal{I}} \\ 0 & B_{\mathrm{st}} \end{pmatrix} \in \mathscr{D}_{m,n}(R)$. Note that

(M1)

$$\begin{aligned}
\widehat{r} * (\mathcal{A} + \mathcal{B}) &= (r_{\mathrm{st}} + r_{\mathcal{I}}\varepsilon) * \left(\begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} + \begin{pmatrix} B_{\mathrm{st}} & B_{\mathcal{I}} \\ 0 & B_{\mathrm{st}} \end{pmatrix} \right) \\
&= (r_{\mathrm{st}} + r_{\mathcal{I}}\varepsilon) * \begin{pmatrix} A_{\mathrm{st}} + B_{\mathrm{st}} & A_{\mathcal{I}} + B_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} + B_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} r_{\mathrm{st}}\left(A_{\mathrm{st}} + B_{\mathrm{st}}\right) & r_{\mathcal{I}}\left(A_{\mathrm{st}} + B_{\mathrm{st}}\right) + r_{\mathrm{st}}\left(A_{\mathcal{I}} + B_{\mathcal{I}}\right) \\ 0 & r_{\mathrm{st}}\left(A_{\mathrm{st}} + B_{\mathrm{st}}\right) \end{pmatrix} \\
&= \begin{pmatrix} r_{\mathrm{st}}A_{\mathrm{st}} + r_{\mathrm{st}}B_{\mathrm{st}} & r_{\mathcal{I}}A_{\mathrm{st}} + r_{\mathcal{I}}B_{\mathrm{st}} + r_{\mathrm{st}}A_{\mathcal{I}} + r_{\mathrm{st}}B_{\mathcal{I}} \\ 0 & r_{\mathrm{st}}A_{\mathrm{st}} + r_{\mathrm{st}}B_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} r_{\mathrm{st}}A_{\mathrm{st}} + r_{\mathrm{st}}B_{\mathrm{st}} & r_{\mathcal{I}}A_{\mathrm{st}} + r_{\mathrm{st}}A_{\mathcal{I}} + r_{\mathcal{I}}B_{\mathrm{st}} + r_{\mathrm{st}}B_{\mathcal{I}} \\ 0 & r_{\mathrm{st}}A_{\mathrm{st}} + r_{\mathrm{st}}B_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} r_{\mathrm{st}}A_{\mathrm{st}} & r_{\mathcal{I}}A_{\mathrm{st}} + r_{\mathrm{st}}A_{\mathcal{I}} \\ 0 & r_{\mathrm{st}}A_{\mathrm{st}} \end{pmatrix} + \begin{pmatrix} r_{\mathrm{st}}B_{\mathrm{st}} & r_{\mathcal{I}}B_{\mathrm{st}} + r_{\mathrm{st}}B_{\mathcal{I}} \\ 0 & r_{\mathrm{st}}B_{\mathrm{st}} \end{pmatrix} \\
&= (r_{\mathrm{st}} + r_{\mathcal{I}}\varepsilon) * \begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} + (r_{\mathrm{st}} + r_{\mathcal{I}}\varepsilon) * \begin{pmatrix} B_{\mathrm{st}} & B_{\mathcal{I}} \\ 0 & B_{\mathrm{st}} \end{pmatrix} \\
&= \widehat{r} * \mathcal{A} + \widehat{r} * \mathcal{B};
\end{aligned}$$

(M2)

$$
\begin{aligned}
(\widehat{r}+\widehat{s})*\mathcal{A} &= (r_{\mathrm{st}}+r_{\mathcal{I}}\varepsilon+s_{\mathrm{st}}+s_{\mathcal{I}}\varepsilon)*\begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} \\
&= (r_{\mathrm{st}}+s_{\mathrm{st}}+(r_{\mathcal{I}}+s_{\mathcal{I}})\,\varepsilon)*\begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} (r_{\mathrm{st}}+s_{\mathrm{st}})\,A_{\mathrm{st}} & (r_{\mathcal{I}}+s_{\mathcal{I}})\,A_{\mathrm{st}}+(r_{\mathrm{st}}+s_{\mathrm{st}})\,A_{\mathcal{I}} \\ 0 & (r_{\mathrm{st}}+s_{\mathrm{st}})\,A_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} r_{\mathrm{st}}A_{\mathrm{st}}+s_{\mathrm{st}}A_{\mathrm{st}} & r_{\mathcal{I}}A_{\mathrm{st}}+s_{\mathcal{I}}A_{\mathrm{st}}+r_{\mathrm{st}}A_{\mathcal{I}}+s_{\mathrm{st}}A_{\mathcal{I}} \\ 0 & r_{\mathrm{st}}A_{\mathrm{st}}+s_{\mathrm{st}}A_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} r_{\mathrm{st}}A_{\mathrm{st}}+s_{\mathrm{st}}A_{\mathrm{st}} & r_{\mathcal{I}}A_{\mathrm{st}}+r_{\mathrm{st}}A_{\mathcal{I}}+s_{\mathcal{I}}A_{\mathrm{st}}+s_{\mathrm{st}}A_{\mathcal{I}} \\ 0 & r_{\mathrm{st}}A_{\mathrm{st}}+s_{\mathrm{st}}A_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} r_{\mathrm{st}}A_{\mathrm{st}} & r_{\mathcal{I}}A_{\mathrm{st}}+r_{\mathrm{st}}A_{\mathcal{I}} \\ 0 & r_{\mathrm{st}}A_{\mathrm{st}} \end{pmatrix}+\begin{pmatrix} s_{\mathrm{st}}A_{\mathrm{st}} & s_{\mathcal{I}}A_{\mathrm{st}}+s_{\mathrm{st}}A_{\mathcal{I}} \\ 0 & s_{\mathrm{st}}A_{\mathrm{st}} \end{pmatrix} \\
&= (r_{\mathrm{st}}+r_{\mathcal{I}}\varepsilon)*\begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix}+(s_{\mathrm{st}}+s_{\mathcal{I}}\varepsilon)*\begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} \\
&= \widehat{r}*\mathcal{A}+\widehat{s}*\mathcal{A};
\end{aligned}
$$

and

(M3)

$$
\begin{aligned}
\widehat{r}*(\widehat{s}*\mathcal{A}) &= (r_{\mathrm{st}}+r_{\mathcal{I}}\varepsilon)*\left((s_{\mathrm{st}}+s_{\mathcal{I}}\varepsilon)*\begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix}\right) \\
&= (r_{\mathrm{st}}+r_{\mathcal{I}}\varepsilon)*\begin{pmatrix} s_{\mathrm{st}}A_{\mathrm{st}} & s_{\mathcal{I}}A_{\mathrm{st}}+s_{\mathrm{st}}A_{\mathcal{I}} \\ 0 & s_{\mathrm{st}}A_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathrm{st}} & r_{\mathcal{I}}s_{\mathrm{st}}A_{\mathrm{st}}+r_{\mathrm{st}}\left(s_{\mathcal{I}}A_{\mathrm{st}}+s_{\mathrm{st}}A_{\mathcal{I}}\right) \\ 0 & r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathrm{st}} & r_{\mathcal{I}}s_{\mathrm{st}}A_{\mathrm{st}}+r_{\mathrm{st}}s_{\mathcal{I}}A_{\mathrm{st}}+r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathcal{I}} \\ 0 & r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathrm{st}} & r_{\mathrm{st}}s_{\mathcal{I}}A_{\mathrm{st}}+r_{\mathcal{I}}s_{\mathrm{st}}A_{\mathrm{st}}+r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathcal{I}} \\ 0 & r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathrm{st}} & (r_{\mathrm{st}}s_{\mathcal{I}}+r_{\mathcal{I}}s_{\mathrm{st}})\,A_{\mathrm{st}}+r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathcal{I}} \\ 0 & r_{\mathrm{st}}s_{\mathrm{st}}A_{\mathrm{st}} \end{pmatrix} \\
&= (r_{\mathrm{st}}s_{\mathrm{st}}+(r_{\mathrm{st}}s_{\mathcal{I}}+r_{\mathcal{I}}s_{\mathrm{st}})\,\varepsilon)*\begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix}
\end{aligned}
$$

$$= ((r_{\text{st}} + r_{\mathcal{I}}\varepsilon)(s_{\text{st}} + s_{\mathcal{I}}\varepsilon)) * \begin{pmatrix} A_{\text{st}} & A_{\mathcal{I}} \\ 0 & A_{\text{st}} \end{pmatrix}$$

$$= (\widehat{rs}) * \mathcal{A}.$$

Hence, $\mathscr{D}_{m,n}(R)$ has a $\mathbb{D}R$-module structure.

Similar to Proposition 2.7, we can also show that $\mathscr{D}_{m,n}(R)$ is a unital $\mathbb{D}R-$module if R is a ring with unity. □

Proposition 2.13. *Let R be a ring. Define $\mathscr{D}_n(R) := \mathscr{D}_{n,n}(R)$. Then, $\mathscr{D}_n(R)$ is a ring under the usual block matrix addition and multiplication.*

Proof. As previously shown, $\mathscr{D}_n(R)$ is nonempty. By Proposition 2.11, $\mathscr{D}_n(R)$ is an additive abelian group. Furthermore,

$$\mathcal{AB} = \begin{pmatrix} A_{\text{st}} & A_{\mathcal{I}} \\ 0 & A_{\text{st}} \end{pmatrix} \begin{pmatrix} B_{\text{st}} & B_{\mathcal{I}} \\ 0 & B_{\text{st}} \end{pmatrix} = \begin{pmatrix} A_{\text{st}}B_{\text{st}} & A_{\text{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\text{st}} \\ 0 & A_{\text{st}}B_{\text{st}} \end{pmatrix} \in \mathscr{D}_n(R)$$

since $M_n(R)$ is closed under matrix multiplication. By the subring test, $\mathscr{D}_n(R)$ is a subring of $M_{2n}(R)$, and so $\mathscr{D}_n(R)$ is a ring under block matrix addition and multiplication. □

Proposition 2.14. *Let R be a commutative ring with unity. Then $\mathscr{D}_n(R)$ has a $\mathbb{D}R$-algebra structure.*

Proof. Since $\mathbb{D}R$ is a commutative ring with unity, then by Proposition 2.13, $\mathscr{D}_n(R)$ is a ring. Note that

(A1) $(\mathscr{D}_n(R), +)$ is a unital $\mathbb{D}R-$module (by Proposition 2.12); and

(A2) for all $\widehat{k} \in \mathbb{D}R$ and $\mathcal{A} = \begin{pmatrix} A_{\text{st}} & A_{\mathcal{I}} \\ 0 & A_{\text{st}} \end{pmatrix}, \mathcal{B} = \begin{pmatrix} B_{\text{st}} & B_{\mathcal{I}} \\ 0 & B_{\text{st}} \end{pmatrix} \in \mathscr{D}_{m,n}(R)$,

$$\widehat{k} * (\mathcal{AB}) = (k_{\text{st}} + k_{\mathcal{I}}\varepsilon) * \left(\begin{pmatrix} A_{\text{st}} & A_{\mathcal{I}} \\ 0 & A_{\text{st}} \end{pmatrix} \begin{pmatrix} B_{\text{st}} & B_{\mathcal{I}} \\ 0 & B_{\text{st}} \end{pmatrix} \right)$$

$$= (k_{\text{st}} + k_{\mathcal{I}}\varepsilon) * \begin{pmatrix} A_{\text{st}}B_{\text{st}} & A_{\text{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\text{st}} \\ 0 & A_{\text{st}}B_{\text{st}} \end{pmatrix}$$

$$= \begin{pmatrix} k_{\text{st}}A_{\text{st}}B_{\text{st}} & k_{\mathcal{I}}A_{\text{st}}B_{\text{st}} + k_{\text{st}}(A_{\text{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\text{st}}) \\ 0 & k_{\text{st}}A_{\text{st}}B_{\text{st}} \end{pmatrix}$$

$$= \begin{pmatrix} k_{\text{st}}A_{\text{st}}B_{\text{st}} & k_{\mathcal{I}}A_{\text{st}}B_{\text{st}} + (k_{\text{st}}A_{\text{st}}B_{\mathcal{I}} + k_{\text{st}}A_{\mathcal{I}}B_{\text{st}}) \\ 0 & k_{\text{st}}A_{\text{st}}B_{\text{st}} \end{pmatrix}$$

$$
\begin{aligned}
&= \begin{pmatrix} k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathrm{st}} & k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathcal{I}} + (k_{\mathcal{I}}A_{\mathrm{st}}B_{\mathrm{st}} + k_{\mathrm{st}}A_{\mathcal{I}}B_{\mathrm{st}}) \\ 0 & k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathrm{st}} & k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathcal{I}} + (k_{\mathcal{I}}A_{\mathrm{st}} + k_{\mathrm{st}}A_{\mathcal{I}})\, B_{\mathrm{st}} \\ 0 & k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} k_{\mathrm{st}}A_{\mathrm{st}} & k_{\mathcal{I}}A_{\mathrm{st}} + k_{\mathrm{st}}A_{\mathcal{I}} \\ 0 & k_{\mathrm{st}}A_{\mathrm{st}} \end{pmatrix} \begin{pmatrix} B_{\mathrm{st}} & B_{\mathcal{I}} \\ 0 & B_{\mathrm{st}} \end{pmatrix} \\
&= \left((k_{\mathrm{st}} + k_{\mathcal{I}}\varepsilon) * \begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} \right) \begin{pmatrix} B_{\mathrm{st}} & B_{\mathcal{I}} \\ 0 & B_{\mathrm{st}} \end{pmatrix} \\
&= \left(\widehat{k} * \mathcal{A} \right) \mathcal{B} \\
&= \begin{pmatrix} k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathrm{st}} & (k_{\mathcal{I}}A_{\mathrm{st}}B_{\mathrm{st}} + k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathcal{I}}) + k_{\mathrm{st}}A_{\mathcal{I}}B_{\mathrm{st}} \\ 0 & k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathrm{st}} & A_{\mathrm{st}}\left(k_{\mathcal{I}}B_{\mathrm{st}} + k_{\mathrm{st}}B_{\mathcal{I}}\right) + k_{\mathrm{st}}A_{\mathcal{I}}B_{\mathrm{st}} \\ 0 & k_{\mathrm{st}}A_{\mathrm{st}}B_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} \begin{pmatrix} k_{\mathrm{st}}B_{\mathrm{st}} & k_{\mathcal{I}}B_{\mathrm{st}} + k_{\mathrm{st}}B_{\mathcal{I}} \\ 0 & k_{\mathrm{st}}B_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} \left((k_{\mathrm{st}} + k_{\mathcal{I}}\varepsilon) * \begin{pmatrix} B_{\mathrm{st}} & B_{\mathcal{I}} \\ 0 & B_{\mathrm{st}} \end{pmatrix} \right) \\
&= \mathcal{A} \left(\widehat{k} * \mathcal{B} \right).
\end{aligned}
$$

Hence, $\mathscr{D}_n(R)$ has a $\mathbb{D}R$-algebra structure. □

Definition 2.15. [8] Let A and B be modules over a ring R. A function $f : A \to B$ is an $R-$*module homomorphism* provided that for all $a, c \in A$ and $r \in R$:

$$f(a + c) = f(a) + f(c) \text{ and } f(ra) = rf(a).$$

Moreover, if f is bijective, then f is said to be an $R-$*module isomorphism* .

Definition 2.16. [8] Let K be a commutative ring with unity and A, B be $K-$algebras. A *homomorphism* (*isomorphism*) *of* K*-algebras* $f : A \to B$ is a ring homomorphism (isomorphism) that is also a $K-$module homomorphism (isomorphism).

Theorem 2.17. *Let* R *be ring. Define* $\phi : M_{m,n}(\mathbb{D}R) \to \mathscr{D}_{m,n}(R)$ *where*

$$\phi(\widehat{A}) = \begin{pmatrix} A_{st} & A_{\mathcal{I}} \\ 0 & A_{st} \end{pmatrix}$$

for all $\widehat{A} \in M_{m,n}(\mathbb{D}R)$. Then, ϕ is a $\mathbb{D}R$-module isomorphism. Moreover, if R is a commutative ring with unity and $m = n$, then ϕ is a $\mathbb{D}R$-algebra isomorphism.

Proof. By Propositions 2.7 and 2.12, both $M_{m,n}(\mathbb{D}R)$ and $\mathscr{D}_{m,n}(R)$ are $\mathbb{D}R$-modules. Moreover, let $\widehat{A}, \widehat{B} \in M_{m,n}(\mathbb{D}R)$ and suppose that $\widehat{A} = \widehat{B}$. Then by equality of dual matrices over R, $A_{\mathrm{st}} = B_{\mathrm{st}}$ and $A_{\mathcal{I}} = B_{\mathcal{I}}$, and so

$$\phi(\widehat{A}) = \phi\left(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}}\right) = \begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} = \begin{pmatrix} B_{\mathrm{st}} & B_{\mathcal{I}} \\ 0 & B_{\mathrm{st}} \end{pmatrix} = \phi\left(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}\right) = \phi(\widehat{B}).$$

Hence, ϕ is a well-defined function.

(I1) Suppose that $\phi(\widehat{A}) = \phi(\widehat{B})$. Then,

$$\phi\left(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}}\right) = \begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} = \begin{pmatrix} B_{\mathrm{st}} & B_{\mathcal{I}} \\ 0 & B_{\mathrm{st}} \end{pmatrix} = \phi\left(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}\right),$$

i.e., $A_{\mathrm{st}} = B_{\mathrm{st}}$ and $A_{\mathcal{I}} = B_{\mathcal{I}}$. By equality of dual matrices over R, $\widehat{A} = \widehat{B}$. Hence, ϕ is injective.

(I2) It is easy to see from the definition that ϕ is surjective.

By (I1) and (I2), ϕ is bijective. Furthermore, ϕ is a $\mathbb{D}R$-module homomorphism. Indeed,

$$\begin{aligned}
\phi\left(\widehat{A} + \widehat{B}\right) &= \phi\left(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}} + B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}\right) \\
&= \phi\left(A_{\mathrm{st}} + B_{\mathrm{st}} + \varepsilon\left(A_{\mathcal{I}} + B_{\mathcal{I}}\right)\right) \\
&= \begin{pmatrix} A_{\mathrm{st}} + B_{\mathrm{st}} & A_{\mathcal{I}} + B_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} + B_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} + \begin{pmatrix} B_{\mathrm{st}} & B_{\mathcal{I}} \\ 0 & B_{\mathrm{st}} \end{pmatrix} \\
&= \phi\left(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}}\right) + \phi\left(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}\right) \\
&= \phi(\widehat{A}) + \phi(\widehat{B})
\end{aligned}$$

and

$$\begin{aligned}
\phi\left(\widehat{r}\widehat{A}\right) &= \phi\left(\left(r_{\mathrm{st}} + r_{\mathcal{I}}\varepsilon\right)\left(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}}\right)\right) \\
&= \phi\left(r_{\mathrm{st}}A_{\mathrm{st}} + \varepsilon\left(r_{\mathrm{st}}A_{\mathcal{I}} + r_{\mathcal{I}}A_{\mathrm{st}}\right)\right) \\
&= \begin{pmatrix} r_{\mathrm{st}}A_{\mathrm{st}} & r_{\mathrm{st}}A_{\mathcal{I}} + r_{\mathcal{I}}A_{\mathrm{st}} \\ 0 & r_{\mathrm{st}}A_{\mathrm{st}} \end{pmatrix}
\end{aligned}$$

$$
\begin{aligned}
&= \begin{pmatrix} r_{\mathrm{st}}A_{\mathrm{st}} & r_{\mathcal{I}}A_{\mathrm{st}} + r_{\mathrm{st}}A_{\mathcal{I}} \\ 0 & r_{\mathrm{st}}A_{\mathrm{st}} \end{pmatrix} \\
&= (r_{\mathrm{st}} + r_{\mathcal{I}}\varepsilon) * \begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix} \\
&= (r_{\mathrm{st}} + r_{\mathcal{I}}\varepsilon) * \phi\left(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}}\right) \\
&= \widehat{r} * \phi\left(\widehat{A}\right)
\end{aligned}
$$

for all $\widehat{r} \in \mathbb{D}R$ and $\widehat{A}, \widehat{B} \in M_{m,n}(\mathbb{D}R)$.
Now, suppose that $m = n$. By Propositions 2.10 and 2.14, $M_n(\mathbb{D}R)$ and $\mathscr{D}_n(R)$ are both $\mathbb{D}R$-algebras. Moreover, by Propositions 2.8 and 2.13, $M_n(\mathbb{D}R)$ and $\mathscr{D}_n(R)$ are rings. Since $\phi(\widehat{A} + \widehat{B}) = \phi(\widehat{A}) + \phi(\widehat{B})$ and

$$
\begin{aligned}
\phi\left(\widehat{A}\widehat{B}\right) &= \phi\left(\left(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}}\right)\left(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}\right)\right) \\
&= \phi\left(A_{\mathrm{st}}B_{\mathrm{st}} + \varepsilon\left(A_{\mathrm{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\mathrm{st}}\right)\right) \\
&= \begin{pmatrix} A_{\mathrm{st}}B_{\mathrm{st}} & A_{\mathrm{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\mathrm{st}} \\ 0 & A_{\mathrm{st}}B_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix}\begin{pmatrix} B_{\mathrm{st}} & B_{\mathcal{I}} \\ 0 & B_{\mathrm{st}} \end{pmatrix} \\
&= \phi\left(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}}\right)\phi\left(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}\right) \\
&= \phi\left(\widehat{A}\right)\phi\left(\widehat{B}\right)
\end{aligned}
$$

for all $\widehat{A}, \widehat{B} \in M_n(\mathbb{D}R)$, ϕ is a ring isomorphism. Therefore, ϕ is an isomorphism of $\mathbb{D}R$-algebras. □

Definition 2.18. Let R be a ring with unity and let $\widehat{A} \in M_n(\mathbb{D}R)$. Then we say that $\widehat{A}$ is *invertible* if there exists $\widehat{B} \in M_n(\mathbb{D}R)$ such that $\widehat{A}\widehat{B} = I_n = \widehat{B}\widehat{A}$.

Remark 2.19.

1. Similar to Proposition 1.7, it can be shown that $\widehat{A} \in M_n(\mathbb{D}R)$ is invertible if and only if A_{st} is invertible, and that its inverse is unique. By this, we denote $\widehat{A}^{-1} \in M_n(\mathbb{D}R)$ to be the inverse of $\widehat{A}$ and is given by $\widehat{A}^{-1} = A_{\mathrm{st}}^{-1} - \varepsilon A_{\mathrm{st}}^{-1}A_{\mathcal{I}}A_{\mathrm{st}}^{-1}$.

2. Note that $\widehat{A}$ is invertible if and only if $\phi(\widehat{A})$ is invertible. Furthermore, $\phi(\widehat{A})^{-1} = \phi(\widehat{A}^{-1})$.

3. Note that $\phi\left(\widehat{A}\widehat{B}\right) = \phi\left(\widehat{A}\right)\phi\left(\widehat{B}\right)$ even though $\widehat{A}$ and $\widehat{B}$ do not necessarily have the same dimensions. Indeed, let $\widehat{A} = A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}} \in M_{m,n}(\mathbb{D}R)$ and $\widehat{B} = B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}} \in M_{n,p}(\mathbb{D}R)$. Then,

$$\begin{aligned}
\phi\left(\widehat{A}\widehat{B}\right) &= \phi\left(\left(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}}\right)\left(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}\right)\right) \\
&= \phi\left(A_{\mathrm{st}}B_{\mathrm{st}} + \varepsilon\left(A_{\mathrm{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\mathrm{st}}\right)\right) \\
&= \begin{pmatrix} A_{\mathrm{st}}B_{\mathrm{st}} & A_{\mathrm{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\mathrm{st}} \\ 0 & A_{\mathrm{st}}B_{\mathrm{st}} \end{pmatrix} \\
&= \begin{pmatrix} A_{\mathrm{st}} & A_{\mathcal{I}} \\ 0 & A_{\mathrm{st}} \end{pmatrix}\begin{pmatrix} B_{\mathrm{st}} & B_{\mathcal{I}} \\ 0 & B_{\mathrm{st}} \end{pmatrix} \\
&= \phi\left(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}}\right)\phi\left(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}\right) \\
&= \phi\left(\widehat{A}\right)\phi\left(\widehat{B}\right).
\end{aligned}$$

Chapter 3

Schur's Triangularization Theorem for Dual Complex Matrices

Schur's triangularization theorem states that any matrix $A \in M_n(\mathbb{C})$ is unitarily similar to an upper triangular matrix $T \in M_n(\mathbb{C})$, i.e., there exists a unitary $U \in M_n(\mathbb{C})$ such that $U^*AU = T = [t_{ij}]$ is upper triangular with diagonal entries $t_{ii} = \lambda_i$, where λ_i is an eigenvalue of A for all $i \in \{1, 2, \ldots, n\}$ [7]. In this chapter, we are interested on how this theorem will be extended to dual complex matrices. The following definition can be found in [14].

Definition 3.1. Let $\widehat{A} \in M_n(\mathbb{DC})$. If there exists $\widehat{\lambda} = \lambda_{\text{st}} + \lambda_{\mathcal{I}}\varepsilon \in \mathbb{DC}$ and an appreciable $\widehat{\mathbf{x}} = \mathbf{x}_{\text{st}} + \varepsilon\mathbf{x}_{\mathcal{I}} \in \mathbb{DC}^n$ satisfying the *eigenvalue-eigenvector equation* $\widehat{A}\widehat{\mathbf{x}} = \widehat{\lambda}\widehat{\mathbf{x}}$, then $\widehat{\lambda}$ is called an *eigenvalue* of $\widehat{A}$ and $\widehat{\mathbf{x}}$ is called an *eigenvector* of $\widehat{A}$ corresponding to $\widehat{\lambda}$ (alternatively, $\widehat{\lambda}$ is an eigenvalue of $\widehat{A}$ corresponding to $\widehat{\mathbf{x}}$). The ordered pair $(\widehat{\lambda}, \widehat{\mathbf{x}})$ is called an *eigenpair* of $\widehat{A}$. Moreover, the set of all eigenvalues of $\widehat{A}$ is called the *spectrum* of $\widehat{A}$ which we denote as $\sigma(\widehat{A})$.

Example 3.2. Given $\widehat{\mu} = \mu_{\text{st}} + \varepsilon\mu_{\mathcal{I}} \in \mathbb{DC}$ and $\widehat{A} \in M_n(\mathbb{DC})$, where $A_{\text{st}} = \mu_{\text{st}}I_n$ and $A_{\mathcal{I}} \in M_n(\mathbb{C})$ is arbitrary, find $\sigma(\widehat{A})$.

Solution. Consider the equation $\widehat{A}\widehat{\mathbf{x}} = \widehat{\lambda}\widehat{\mathbf{x}}$, where $\widehat{\mathbf{x}} \in \mathbb{DC}^n$ is appreciable and $\widehat{\lambda} \in \mathbb{DC}$. Then, we have the following system of equations:

$$A_{\text{st}}\mathbf{x}_{\text{st}} = \lambda_{\text{st}}\mathbf{x}_{\text{st}} \tag{3.1}$$

$$A_{\text{st}}\mathbf{x}_{\mathcal{I}} + A_{\mathcal{I}}\mathbf{x}_{\text{st}} = \lambda_{\text{st}}\mathbf{x}_{\mathcal{I}} + \lambda_{\mathcal{I}}\mathbf{x}_{\text{st}} \tag{3.2}$$

From (3.1), $\lambda_{\text{st}} \in \sigma(A_{\text{st}})$, and so $\lambda_{\text{st}} = \mu_{\text{st}}$. In particular from (3.2), $A_{\mathcal{I}}\mathbf{x}_{\text{st}} = \lambda_{\mathcal{I}}\mathbf{x}_{\text{st}}$. Thus, $\sigma(\widehat{A}) \subseteq \{\widehat{\lambda} \in \mathbb{DC} \mid \lambda_{\text{st}} = \mu_{\text{st}} \text{ and } \lambda_{\mathcal{I}} \in \sigma(A_{\mathcal{I}})\}$. The reverse inclusion can be seen by substituting $\lambda_{\text{st}} = \mu_{\text{st}}$, $\lambda_{\mathcal{I}} \in \sigma(A_{\mathcal{I}})$, and $\widehat{\mathbf{x}} = e_i$ to the eigenvector-eigenvector equation of $\widehat{A}$, where $e_i \in \mathbb{C}^n$ denotes the ith standard unit basis vector.

Example 3.3. Given $\widehat{\mu} = \mu_{\text{st}} + \varepsilon\mu_{\mathcal{I}} \in \mathbb{DC}$ and $\widehat{A} \in M_n(\mathbb{DC})$, where

$$A_{\text{st}} = \begin{pmatrix} \mu_{\text{st}} & 1 & \dots & 0 \\ 0 & \mu_{\text{st}} & \ddots & \vdots \\ \vdots & \vdots & \ddots & 1 \\ 0 & 0 & \dots & \mu_{\text{st}} \end{pmatrix}$$

is an upper triangular matrix with all μ_{st} on its main diagonal and all 1 on its superdiagonal. What are necessary and sufficient conditions on $A_{\mathcal{I}}$ so that $\sigma(\widehat{A})$ is nonempty? For now, we will withhold the computation of eigenvalues of $\widehat{A}$ in this example until the discussion of Remark 3.15.

Definition 3.4 (Conjugate Transpose)**.** [7] Let $A := [a_{ij}] \in M_{m,n}(\mathbb{C})$. The conjugate transpose of A is denoted by A^* and is defined by $A^* = \overline{A}^T$, where $\overline{A}$ is the entrywise conjugate of A.

Definition 3.5. [7] Let $A \in M_n(\mathbb{C})$. We say that
(a) A is *Hermitian* if $A^* = A$;
(b) A is *skew-Hermitian* if $A^* = -A$;
(c) A is *normal* if $AA^* = A^*A$; and
(d) A is *unitary* if $AA^* = A^*A = I_n$.

Definition 3.6 (Moore-Penrose Inverse)**.** [7] Let $A \in M_{m,n}(\mathbb{C})$. If $X \in M_{n,m}(\mathbb{C})$ satisfies the following *Moore-Penrose* (MP) *conditions:*

(P1) $AXA = A$;

(P2) $XAX = X$;

(P3) $(AX)^* = AX$; and

(P4) $(XA)^* = XA$,

then X is a *Moore-Penrose inverse* (MPI) of A.

Proposition 3.7. *[12] Let $A \in M_{m,n}(\mathbb{C})$. Then the MPI of A is unique.*

Proof. Let $X \in M_{n,m}(\mathbb{C})$ and $Y \in M_{n,m}(\mathbb{C})$ be Moore-Penrose inverses of A. Then

$$\begin{aligned} X &= X(AX) \\ &= XX^*A^* \\ &= XX^*A^*(Y^*A^*) \end{aligned}$$

$$
\begin{aligned}
&= X(X^*A^*)AY \\
&= X(AXA)Y \\
&= XAY \\
&= XA(YA)Y \\
&= (XA)A^*Y^*Y \\
&= (A^*X^*A^*)Y^*Y \\
&= (A^*Y^*)Y \\
&= YAY \\
&= Y.
\end{aligned}
$$

Thus, the MPI of A is unique. □

Remark 3.8.

1. We denote $A^\dagger \in M_{n,m}(\mathbb{C})$ to be *the* MPI of A.
2. Note that the MPI of the MPI of A is A itself, i.e., $(A^\dagger)^\dagger = A$.
3. For an arbitrary invertible matrix $A \in M_n(\mathbb{C})$, its inverse $A^{-1} \in M_n(\mathbb{C})$ is the MPI of A.
4. If $m = n = 1$, then the MPI of $a \in \mathbb{C}$ is $a^\dagger = a^{-1}$ if a is nonzero and $a^\dagger = 0$ otherwise.

Theorem 3.9 (Singular Value Decomposition for Matrices over $\mathbb{C}$)**.** *[7] Let $A \in M_{m,n}(\mathbb{C})$, let $q = min\{m, n\}$, and suppose rank $A = r$. Then, there are unitary matrices $V \in M_m(\mathbb{C})$ and $W \in M_n(\mathbb{C})$ and a square diagonal matrix*

$$\Sigma_q = \begin{pmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_q \end{pmatrix} \tag{3.3}$$

such that $\sigma_i \in \mathbb{C}$ for all $i \in \{1, 2, \dots, r, r+1, \dots q\}$ and $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0 = \sigma_{r+1} = \dots = \sigma_q$ and $A = V\Sigma W^*$, in which

$$\Sigma = \begin{cases} \Sigma_q & \text{if } m = n \\ \begin{pmatrix} \Sigma_q & 0_{m,n-m} \end{pmatrix} \in M_{m,n}(\mathbb{C}) & \text{if } m < n, \text{ and} \\ \begin{pmatrix} \Sigma_q \\ 0_{m-n,n} \end{pmatrix} \in M_{m,n}(\mathbb{C}) & \text{if } m > n. \end{cases}$$

This theorem implies that a singular value decomposition (SVD) exists for all matrices over $\mathbb{C}$. It turns out, we can use SVD to compute the MPI of an arbitrary matrix $A \in M_{m,n}(\mathbb{C})$.

Proposition 3.10. *Let Σ_q be the square diagonal matrix in* (3.3)*. Then the MPI of Σ_q is given by* $\Sigma_q^\dagger = \begin{pmatrix} \sigma_1^\dagger & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_q^\dagger \end{pmatrix}$.

Proof. It suffices to show that $\Sigma_q^\dagger$ satisfies the four MP conditions with respect to Σ :

(P1)

$$\begin{aligned} e_i^T \Sigma_q \Sigma_q^\dagger \Sigma_q e_j &= \begin{pmatrix} 0 & 0 & \dots & \sigma_i \sigma_i^\dagger & \dots & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & \dots & \sigma_j & \dots & 0 \end{pmatrix}^T \\ &= \begin{cases} 0, & \text{if } i \neq j \\ \sigma_i, & \text{if } i = j \end{cases} \\ &= e_i^T \Sigma_q e_j; \end{aligned}$$

(P2)

$$\begin{aligned} e_i^T \Sigma_q^\dagger \Sigma_q \Sigma_q^\dagger e_j &= \begin{pmatrix} 0 & 0 & \dots & \sigma_i^\dagger \sigma_i & \dots & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & \dots & \sigma_j^\dagger & \dots & 0 \end{pmatrix}^T \\ &= \begin{cases} 0, & \text{if } i \neq j \\ \sigma_i^\dagger, & \text{if } i = j \end{cases} \\ &= e_i^T \Sigma_q^\dagger e_j; \end{aligned}$$

(P3) Since $\mathbb{C}$ is commutative and Σ_q and $\Sigma_q^\dagger$ are both diagonal matrices over $\mathbb{C}$, they commute with one another. Hence, $(\Sigma_q \Sigma_q^\dagger)^T = (\Sigma_q^\dagger)^T {\Sigma_q}^T = \Sigma_q^\dagger \Sigma_q = \Sigma_q \Sigma_q^\dagger$; and

(P4) $(\Sigma_q^\dagger \Sigma_q)^T = \Sigma_q^T (\Sigma_q^\dagger)^T = \Sigma_q \Sigma_q^\dagger = \Sigma_q^\dagger \Sigma_q$.

Thus, $\Sigma_q^\dagger$ is indeed the MPI of Σ_q. □

Proposition 3.11. *Let* $\Sigma = \begin{pmatrix} \Sigma_q & 0_{n,m-n} \end{pmatrix}^T$ *or* $\Sigma = \begin{pmatrix} \Sigma_q & 0_{m,n-m} \end{pmatrix}$*, where Σ_q is the square diagonal matrix in* (3.3)*. Then the MPI of Σ is given by* $\Sigma^\dagger = \begin{pmatrix} \Sigma_q^\dagger & 0_{n,m-n} \end{pmatrix}$ *or* $\Sigma^\dagger = \begin{pmatrix} \Sigma_q^\dagger & 0_{m,n-m} \end{pmatrix}^T$*, respectively.*

Proof. Suppose $\Sigma = \begin{pmatrix} \Sigma_q & 0_{n,m-n} \end{pmatrix}^T$. We need to show that $\Sigma^\dagger$ satisfies the four MP conditions with respect to Σ :

(P1) $\Sigma\Sigma^\dagger\Sigma = \begin{pmatrix} \Sigma_q\Sigma_q^\dagger & 0_{n,m-n} \\ 0_{m-n,n} & 0_{m-n,m-n} \end{pmatrix} \begin{pmatrix} \Sigma_q \\ 0_{m-n,n} \end{pmatrix} = \begin{pmatrix} \Sigma_q\Sigma_q^\dagger\Sigma_q \\ 0_{m-n,n} \end{pmatrix} = \Sigma;$

(P2) $\Sigma^\dagger\Sigma\Sigma^\dagger = \begin{pmatrix} \Sigma_q^\dagger\Sigma_q \end{pmatrix} \begin{pmatrix} \Sigma_q^\dagger & 0_{n,m-n} \end{pmatrix} = \begin{pmatrix} \Sigma_q^\dagger\Sigma_q\Sigma_q^\dagger \\ 0_{m-n,n} \end{pmatrix}^T = \Sigma^\dagger;$

(P3) $\Sigma\Sigma^\dagger = \begin{pmatrix} \Sigma_q\Sigma_q^\dagger & 0_{n,m-n} \\ 0_{m-n,n} & 0_{m-n,m-n} \end{pmatrix}$ is Hermitian since $\Sigma_q\Sigma_q^\dagger$ is Hermitian; and

(P4) $\Sigma^\dagger\Sigma = \Sigma_q^\dagger\Sigma_q$ is Hermitian since $\Sigma_q^\dagger\Sigma_q$ is Hermitian.

In a similar fashion, we can show that $\Sigma^\dagger = \begin{pmatrix} \Sigma_q^\dagger & 0_{m,n-m} \end{pmatrix}^T$ if $\Sigma = \begin{pmatrix} \Sigma_q & 0_{m,n-m} \end{pmatrix}$. □

Corollary 3.12. *[7] Let $A = V\Sigma W^*$ be an SVD of $A \in M_{m,n}(\mathbb{C})$, where $V \in M_m(\mathbb{C})$ and $W \in M_n(\mathbb{C})$ are both unitary matrices and $\Sigma \in M_{m,n}(\mathbb{C})$ is as defined in Theorem 3.9. Then $A^\dagger = W\Sigma^\dagger V^*$ is the MPI of A .*

Proof. It suffices to show that $W\Sigma^\dagger V^*$ satisfies the four MP conditions with respect to A:

(P1) $A(W\Sigma^\dagger V^*)A = (V\Sigma W^*)W\Sigma^\dagger V^*(V\Sigma W^*) = V\Sigma\Sigma^\dagger\Sigma W^* = V\Sigma W^* = A$

(P2) $(W\Sigma^\dagger V^*)A(W\Sigma^\dagger V^*) = W\Sigma^\dagger V^*(V\Sigma W^*)W\Sigma^\dagger V^* = W\Sigma^\dagger\Sigma\Sigma^\dagger V^* = W\Sigma^\dagger V^*$

(P3) $$\begin{aligned}(AW\Sigma^\dagger V^*)^* &= (V\Sigma W^*W\Sigma^\dagger V^*)^* \\ &= (V\Sigma\Sigma^\dagger V^*)^* \\ &= V(\Sigma\Sigma^\dagger)^*V^* \\ &= V\Sigma\Sigma^\dagger V^* \\ &= AW\Sigma^\dagger V^*\end{aligned}$$

(P4) $$\begin{aligned}(W\Sigma^\dagger V^*A)^* &= (W\Sigma^\dagger V^*V\Sigma W^*)^* \\ &= (W\Sigma^\dagger\Sigma W^*)^* \\ &= W(\Sigma^\dagger\Sigma)^*W^* \\ &= W\Sigma^\dagger\Sigma W^* \\ &= W\Sigma^\dagger V^*A.\end{aligned}$$

Thus, $A^\dagger = W\Sigma^\dagger V^*$. □

Theorem 3.13. *[1][2] Consider the equation*

$$AX - YB = C, \tag{3.4}$$

where $A, Y \in M_{n,p}(\mathbb{C})$, $X, B \in M_{p,q}(\mathbb{C})$, *and* $C \in M_{n,q}(\mathbb{C})$ *such that* X *and* Y *are the unknowns. Then* X *and* Y *exist if and only if*

$$(I_n - AA^\dagger)C(I_q - B^\dagger B) = 0. \tag{3.5}$$

If this condition is satisfied, then the general solution is given by

$$\begin{aligned} X &= A^\dagger C + A^\dagger ZB + \left(I_p - A^\dagger A\right) W, \\ Y &= -\left(I_n - AA^\dagger\right) CB^\dagger + Z - \left(I_n - AA^\dagger\right) ZBB^\dagger \end{aligned} \tag{3.6}$$

for any $W \in M_{p,q}(\mathbb{C})$ *and* $Z \in M_{n,p}(\mathbb{C})$.

Proof. ($\Rightarrow$) Multiplying $I_n - AA^\dagger$ from the left of (3.4) yields

$$-\left(I_n - AA^\dagger\right) YB = \left(I_n - AA^\dagger\right) C \tag{3.7}$$

Now, multiplying $I_q - B^\dagger B$ from the right of (3.7) yields

$$0 = \left(I_n - AA^\dagger\right) C \left(I_q - B^\dagger B\right).$$

($\Leftarrow$) Note that we can rewrite (3.5) as

$$AA^\dagger C + \left(CB^\dagger - AA^\dagger CB^\dagger\right) B = C.$$

Thus, a solution to (3.4) exists and is given by $X = A^\dagger C$ and $Y = -\left(I_n - AA^\dagger\right) CB^\dagger$.

It suffices to show that $AX - YB - C = 0$ so that (3.6) satisfies (3.4). Indeed, by substitution we have

$$\begin{aligned} AX - YB - C &= A\left(A^\dagger C + A^\dagger ZB + \left(I_p - A^\dagger A\right) W\right) - \\ &\quad \left(-\left(I_n - AA^\dagger\right) CB^\dagger + Z - \left(I_n - AA^\dagger\right) ZBB^\dagger\right) B - C \\ &= AA^\dagger C + \left(CB^\dagger B - AA^\dagger CB^\dagger B\right) - C \\ &= -\left(I_n - AA^\dagger\right) C \left(I_q - B^\dagger B\right) \\ &= 0. \end{aligned}$$

Thus, (3.6) is a solution to (3.4). Now, suppose that there exists $X_0 \in M_{p,q}(\mathbb{C})$ and $Y_0 \in M_{n,p}(\mathbb{C})$ such that $AX_0 - Y_0 B = C$. Since W and Z are arbitrary, we may take $W = X_0$ and $Z = Y_0$ such that

$$\begin{aligned} X &= A^\dagger C + A^\dagger Y_0 B + \left(I - A^\dagger A\right) X_0 \\ Y &= -\left(I - AA^\dagger\right) CB^\dagger + Y_0 - \left(I - AA^\dagger\right) Y_0 BB^\dagger \end{aligned}$$

is a solution of (3.4). Thus, we have

$$\begin{aligned} X &= A^\dagger C + A^\dagger Y_0 B + \left(I_p - A^\dagger A\right) X_0 \\ &= A^\dagger C + A^\dagger Y_0 B + X_0 - A^\dagger \left(A X_0\right) \\ &= A^\dagger C + A^\dagger Y_0 B + X_0 - A^\dagger \left(C + Y_0 B\right) \\ &= X_0 \end{aligned}$$

and

$$\begin{aligned} Y &= -\left(I_n - AA^\dagger\right) CB^\dagger + Y_0 - \left(I_n - AA^\dagger\right) \left(Y_0 B\right) B^\dagger \\ &= -\left(I_n - AA^\dagger\right) CB^\dagger + Y_0 - \left(I_n - AA^\dagger\right) \left(AX_0 - C\right) B^\dagger \\ &= -\left(I_n - AA^\dagger\right) CB^\dagger + Y_0 - \left(\left(I_n - AA^\dagger\right) AX_0 - \left(I_n - AA^\dagger\right) C\right) B^\dagger \\ &= -\left(I_n - AA^\dagger\right) CB^\dagger + Y_0 + \left(I_n - AA^\dagger\right) CB^\dagger \\ &= Y_0. \end{aligned}$$

Hence, (3.6) is the general solution for (3.4). □

Corollary 3.14. *Let $A \in M_{n,p}(\mathbb{C})$, $X \in M_{p,q}(\mathbb{C})$, and $C \in M_{n,q}(\mathbb{C})$, where X is the unknown. Then the equation $AX = C$ has a solution if and only if*

$$(I_n - AA^\dagger)C = 0.$$

If such solution exists, then the general solution is given by

$$X = A^\dagger C + (I_p - A^\dagger A)W,$$

where $W \in M_{p,q}(\mathbb{C})$ is arbitrary.

Proof. This is a direct consequence of Theorem 3.13, where B is the zero $p \times q$ matrix. □

Remark 3.15. Recall from Example 3.2 that the eigenvalue-eigenvector equation $\widehat{A}\widehat{\mathbf{x}} = \widehat{\lambda}\widehat{\mathbf{x}}$ is equivalent to

$$A_{\mathrm{st}}\mathbf{x}_{\mathrm{st}} = \lambda_{\mathrm{st}}\mathbf{x}_{\mathrm{st}} \tag{3.8}$$

$$A_{\mathrm{st}}\mathbf{x}_{\mathcal{I}} + A_{\mathcal{I}}\mathbf{x}_{\mathrm{st}} = \lambda_{\mathrm{st}}\mathbf{x}_{\mathcal{I}} + \lambda_{\mathcal{I}}\mathbf{x}_{\mathrm{st}}. \tag{3.9}$$

Note that (3.9) is equivalent to

$$\begin{pmatrix} A_{\mathrm{st}} - \lambda_{\mathrm{st}} I_n & \mathbf{x}_{\mathrm{st}} \end{pmatrix} \begin{pmatrix} \mathbf{x}_{\mathcal{I}} \\ \lambda_{\mathcal{I}} \end{pmatrix} = -A_{\mathcal{I}}\mathbf{x}_{\mathrm{st}}. \tag{3.10}$$

By the previous corollary, (3.10) has a solution if and only if

$$\left(I_n - \begin{pmatrix} A_{\mathrm{st}} - \lambda_{\mathrm{st}} I_n & -\mathbf{x}_{\mathrm{st}} \end{pmatrix} \begin{pmatrix} A_{\mathrm{st}} - \lambda_{\mathrm{st}} I_n & -\mathbf{x}_{\mathrm{st}} \end{pmatrix}^\dagger\right) A_{\mathcal{I}}\mathbf{x}_{\mathrm{st}} = 0. \tag{3.11}$$

Now, we are ready to solve Example 3.3 using the concepts discussed above.

Solution (for Example 3.3). From the given, we have $A_{\mathcal{I}} := (b_{ij}) = (\mathbf{b}_1\, \mathbf{b}_2 \ldots \mathbf{b}_n)$, where $\mathbf{b}_j \in \mathbb{C}^n$ is the jth column of $A_{\mathcal{I}}$ for all $j = 1, 2, \ldots, n$. Consider the equation

$$\widehat{A}\widehat{\mathbf{x}} = \widehat{\lambda}\widehat{\mathbf{x}}, \tag{3.12}$$

where $\widehat{\lambda} \in \mathbb{DC}$ and $\widehat{\mathbf{x}} \in \mathbb{DC}^n$ is appreciable. From Remark 3.15 we have

$$A_{\text{st}}\mathbf{x}_{\text{st}} = \lambda_{\text{st}}\mathbf{x}_{\text{st}} \tag{3.13}$$

$$\begin{pmatrix} A_{\text{st}} - \lambda_{\text{st}} I_n & \mathbf{x}_{\text{st}} \end{pmatrix} \begin{pmatrix} \mathbf{x}_{\mathcal{I}} \\ \lambda_{\mathcal{I}} \end{pmatrix} = -A_{\mathcal{I}}\mathbf{x}_{\text{st}}. \tag{3.14}$$

From (3.13), $\lambda_{\text{st}} = \mu_{\text{st}}$ is an eigenvalue of A_{st} corresponding to $\mathbf{x}_{\text{st}} = \begin{pmatrix} -z & 0 & \ldots & 0 \end{pmatrix}^T$, where $z \in \mathbb{C}$ is nonzero. In particular, $A_{\mathcal{I}}\mathbf{x}_{\text{st}} = -z\mathbf{b_1}$. Hence, from Remark 3.15, (3.14) has a solution if and only if

$$\left(I_n - \begin{pmatrix} A_{\text{st}} - \lambda_{\text{st}} I_n & -\mathbf{x}_{\text{st}} \end{pmatrix} \begin{pmatrix} A_{\text{st}} - \lambda_{\text{st}} I_n & -\mathbf{x}_{\text{st}} \end{pmatrix}^{\dagger} \right) A_{\mathcal{I}}\mathbf{x}_{\text{st}} = 0, \tag{3.15}$$

where $\text{M} := \begin{pmatrix} A_{\text{st}} - \lambda_{\text{st}} I_n & -\mathbf{x}_{\text{st}} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & \ldots & 0 & z \\ 0 & 0 & 1 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ddots & 1 & 0 \\ 0 & 0 & 0 & \ldots & 0 & 0 \end{pmatrix} \in M_{n,n+1}(\mathbb{C})$. We claim that

$$\text{M}^{\dagger} = \begin{pmatrix} 0 & 0 & \ldots & 0 & 0 \\ \frac{1}{1+|z|^2} & 0 & \ldots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & 1 & \vdots \\ \frac{\overline{z}}{1+|z|^2} & 0 & \ldots & 0 & 0 \end{pmatrix} \in M_{n+1,n}(\mathbb{C}).$$

Indeed, if $\mathrm{N} := \begin{pmatrix} 0 & 0 & \dots & 0 & 0 \\ \frac{1}{1+|z|^2} & 0 & \dots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & 1 & 0 \\ \frac{\overline{z}}{1+|z|^2} & 0 & \dots & 0 & 0 \end{pmatrix} \in M_{n+1,n}(\mathbb{C})$ then

$$\mathrm{MN} = \begin{pmatrix} \frac{1}{1+|z|^2} + \frac{|z|^2}{1+|z|^2} & 0 & \dots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & 1 & 0 \\ 0 & 0 & \dots & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & 1 & 0 \\ 0 & 0 & \dots & 0 & 0 \end{pmatrix} \in M_n(\mathbb{C}),$$

and so

$$\text{(P1) } \mathrm{MNM} = \begin{pmatrix} 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & 1 & 0 \\ 0 & 0 & \dots & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & \dots & 0 & z \\ 0 & 0 & 1 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ddots & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 \end{pmatrix} = \mathrm{M},$$

$$\text{(P2) } \mathrm{NMN} = \begin{pmatrix} 0 & 0 & \dots & 0 & 0 \\ \frac{1}{1+|z|^2} & 0 & \dots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & 1 & 0 \\ \frac{\overline{z}}{1+|z|^2} & 0 & \dots & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & 1 & 0 \\ 0 & 0 & \dots & 0 & 0 \end{pmatrix} = \mathrm{N},$$

$$\text{(P3) } \mathrm{MN} = \begin{pmatrix} \frac{1}{1+|z|^2} + \frac{|z|^2}{1+|z|^2} & 0 & \dots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & 1 & 0 \\ 0 & 0 & \dots & 0 & 0 \end{pmatrix} \text{ is Hermitian, and}$$

(P4)

$$\mathrm{NM} = \begin{pmatrix} 0 & 0 & \dots & 0 & 0 \\ \frac{1}{1+|z|^2} & 0 & \dots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & 1 & 0 \\ \frac{\overline{z}}{1+|z|^2} & 0 & \dots & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & \dots & 0 & z \\ 0 & 0 & 1 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ddots & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & \frac{1}{1+|z|^2} & 0 & \dots & 0 & \frac{z}{1+|z|^2} \\ 0 & 0 & 1 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ddots & 1 & 0 \\ 0 & \frac{\overline{z}}{1+|z|^2} & 0 & \dots & 0 & \frac{|z|^2}{1+|z|^2} \end{pmatrix}$$

is Hermitian.

Thus, (3.15) is equivalent to

$$\left(I_n - \begin{pmatrix} 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \ddots & 1 & 0 \\ 0 & 0 & \dots & 0 & 0 \end{pmatrix} \right) A_{\mathcal{I}} \mathbf{x}_{\mathrm{st}} = \begin{pmatrix} 0 & \dots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & 0 & 0 \\ 0 & \dots & 0 & 1 \end{pmatrix} A_{\mathcal{I}}(-ze_1) = 0.$$

Since z is nonzero, $e_n^T A_{\mathcal{I}} e_1 = 0$. Therefore, $\widehat{A}$ has a nonempty spectrum if and only if $(n,1)$ entry of $A_{\mathcal{I}}$ is zero. In particular, (3.14) has a solution given by

$$\begin{pmatrix} \mathbf{x}_{\mathcal{I}} \\ \lambda_{\mathcal{I}} \end{pmatrix} = \mathrm{M}^{\dagger}(-A_{\mathcal{I}} \mathbf{x}_{\mathrm{st}}) + \left(I_n - \mathrm{M}^{\dagger}\mathrm{M} \right) W,$$

where $W := \begin{pmatrix} w_1 & w_2 & \dots & w_n \end{pmatrix}^T \in \mathbb{C}^n$ is arbitrary. Note that

$$\begin{aligned} \lambda_I &= e_n^T \begin{pmatrix} \mathbf{x}_{\mathcal{I}} \\ \lambda_{\mathcal{I}} \end{pmatrix} \\ &= \begin{pmatrix} \frac{\overline{z}}{1+|z|^2} & 0 & \dots & 0 \end{pmatrix} \begin{pmatrix} b_{11} \\ b_{21} \\ \vdots \\ b_{n1} \end{pmatrix} (z) + \begin{pmatrix} 0 & \frac{\overline{z}}{1+|z|^2} & 0 & \dots & 0 & \frac{|z|^2}{1+|z|^2} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} \\ &= b_{11} \frac{\overline{z}}{1+|z|^2} + w_2 \frac{\overline{z}}{1+|z|^2} + w_n \frac{1}{1+|z|^2}, \end{aligned}$$

where $A_{\mathcal{I}} := (b_{ij})$. Thus, if we take $w_2 = -b_{11}$ and $w_n = (1+|z|^2)w$, where $w \in \mathbb{C}$ is arbitrary, then $\lambda_{\mathcal{I}} = w \in \mathbb{C}$. Therefore, $\sigma(\widehat{A}) = \{\widehat{\lambda} \in \mathbb{DC} \mid \lambda_{\text{st}} = \mu_{\text{st}}\}$.

Definition 3.16. [14] Let $\widehat{a} \in \mathbb{DC}$ and $\widehat{A} \in M_{m,n}(\mathbb{DC})$. We define the *conjugate* $\widehat{a}^*$ of $\widehat{a}$ as $\widehat{a}^* = \overline{a_{\text{st}}} + \overline{a_{\mathcal{I}}}\varepsilon$. Similarly, we define the *conjugate transpose* $\widehat{A}^*$ of $\widehat{A}$ as $\widehat{A}^* = {A_{\text{st}}}^* + \varepsilon {A_{\mathcal{I}}}^*$.

Theorem 3.17 (Properties of the conjugate transpose)**.** *Let $\widehat{r} = r_{st} + r_{\mathcal{I}}\varepsilon \in \mathbb{DC}$ and $\widehat{A}, \widehat{B} \in M_n(\mathbb{DC})$. Then,*

1. $(\widehat{A}^*)^* = \widehat{A}$
2. $(\widehat{A} \pm \widehat{B})^* = \widehat{A}^* \pm \widehat{B}^*$
3. $(\widehat{r}\widehat{B})^* = \widehat{r}^*\widehat{B}^*$
4. $(\widehat{A}\widehat{B})^* = \widehat{B}^*\widehat{A}^*$
5. $(\widehat{A}^{-1})^* = (\widehat{A}^*)^{-1}$ *if $\widehat{A}$ is invertible*
6. *If $\widehat{\boldsymbol{x}} \in \mathbb{DC}^n$ is appreciable, then $\widehat{\boldsymbol{x}}^*\widehat{\boldsymbol{x}} \in \mathbb{DR}$ and its standard part is strictly positive* [14].

Proof.

1. $(\widehat{A}^*)^* = ({A_{\text{st}}}^* + \varepsilon {A_{\mathcal{I}}}^*)^* = ({A_{\text{st}}}^*)^* + \varepsilon({A_{\mathcal{I}}}^*)^* = A_{\text{st}} + \varepsilon A_{\mathcal{I}} = \widehat{A}$.

2. $$\begin{aligned}(\widehat{A} \pm \widehat{B})^* &= ((A_{\text{st}} + \varepsilon A_{\mathcal{I}}) \pm (B_{\text{st}} + \varepsilon B_{\mathcal{I}}))^* \\ &= (A_{\text{st}} \pm B_{\text{st}} + \varepsilon(A_{\mathcal{I}} \pm B_{\mathcal{I}}))^* \\ &= (A_{\text{st}} \pm B_{\text{st}})^* + \varepsilon(A_{\mathcal{I}} \pm B_{\mathcal{I}})^* \\ &= {A_{\text{st}}}^* + \varepsilon {A_{\mathcal{I}}}^* \pm ({B_{\text{st}}}^* + \varepsilon {B_{\mathcal{I}}}^*) \\ &= \widehat{A}^* \pm \widehat{B}^*.\end{aligned}$$

3. $$\begin{aligned}(\widehat{r}\widehat{A})^* &= ((r_{\text{st}} + \varepsilon r_{\mathcal{I}})(A_{\text{st}} + \varepsilon A_{\mathcal{I}}))^* \\ &= (r_{\text{st}}A_{\text{st}} + \varepsilon(r_{\text{st}}A_{\mathcal{I}} + r_{\mathcal{I}}A_{\text{st}}))^* \\ &= (r_{\text{st}}A_{\text{st}})^* + \varepsilon(r_{\text{st}}A_{\mathcal{I}} + r_{\mathcal{I}}A_{\text{st}})^* \\ &= \overline{r_{\text{st}}}{A_{\text{st}}}^* + \varepsilon((r_{\text{st}}A_{\mathcal{I}})^* + (r_{\mathcal{I}}A_{\text{st}})^*)\end{aligned}$$

$$\begin{aligned}
&= \overline{r_{\mathrm{st}}}{A_{\mathrm{st}}}^* + \varepsilon(\overline{r_{\mathrm{st}}}{A_{\mathcal{I}}}^* + \overline{r_{\mathcal{I}}}{A_{\mathrm{st}}}^*) \\
&= (\overline{r_{\mathrm{st}}} + \varepsilon\overline{r_{\mathcal{I}}})({A_{\mathrm{st}}}^* + \varepsilon{A_{\mathcal{I}}}^*) \\
&= (r_{\mathrm{st}} + \varepsilon r_{\mathcal{I}})^*(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}})^* \\
&= \widehat{r}^*\widehat{A}^*.
\end{aligned}$$

4.

$$\begin{aligned}
(\widehat{A}\widehat{B})^* &= ((A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}})(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}))^* \\
&= (A_{\mathrm{st}}B_{\mathrm{st}} + \varepsilon(A_{\mathrm{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\mathrm{st}}))^* \\
&= (A_{\mathrm{st}}B_{\mathrm{st}})^* + \varepsilon(A_{\mathrm{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\mathrm{st}})^* \\
&= {B_{\mathrm{st}}}^*{A_{\mathrm{st}}}^* + \varepsilon((A_{\mathrm{st}}B_{\mathcal{I}})^* + (A_{\mathcal{I}}B_{\mathrm{st}})^*) \\
&= {B_{\mathrm{st}}}^*{A_{\mathrm{st}}}^* + \varepsilon({B_{\mathcal{I}}}^*{A_{\mathrm{st}}}^* + {B_{\mathrm{st}}}^*{A_{\mathcal{I}}}^*) \\
&= ({B_{\mathrm{st}}}^* + \varepsilon{B_{\mathcal{I}}}^*)({A_{\mathrm{st}}}^* + \varepsilon{A_{\mathcal{I}}}^*) \\
&= (B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}})^*(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}})^* \\
&= \widehat{B}^*\widehat{A}^*.
\end{aligned}$$

5.

$$\begin{aligned}
(\widehat{A}^*)^{-1} &= (A_{\mathrm{st}}^{-1})^* - \varepsilon(A_{\mathrm{st}}^{-1}A_{\mathcal{I}}A_{\mathrm{st}}^{-1})^* \\
&= ({A_{\mathrm{st}}}^*)^{-1} - \varepsilon(A_{\mathrm{st}}^{-1})^*{A_{\mathcal{I}}}^*(A_{\mathrm{st}}^{-1})^* \\
&= ({A_{\mathrm{st}}}^*)^{-1} - \varepsilon({A_{\mathrm{st}}}^*)^{-1}{A_{\mathcal{I}}}^*({A_{\mathrm{st}}}^*)^{-1} \\
&= ({A_{\mathrm{st}}}^* + \varepsilon{A_{\mathcal{I}}}^*)^{-1} \\
&= (\widehat{A}^*)^{-1}.
\end{aligned}$$

6. Suppose $\widehat{\mathbf{x}} \in \mathbb{DC}^n$ is appreciable. Then

$$\begin{aligned}
\widehat{\mathbf{x}}^*\widehat{\mathbf{x}} &= {\mathbf{x}_{\mathrm{st}}}^*\mathbf{x}_{\mathrm{st}} + \varepsilon({\mathbf{x}_{\mathrm{st}}}^*\mathbf{x}_{\mathcal{I}} + {\mathbf{x}_{\mathcal{I}}}^*\mathbf{x}_{\mathrm{st}}) \\
&= \|\mathbf{x}_{\mathrm{st}}\|^2 + \varepsilon({\mathbf{x}_{\mathrm{st}}}^*\mathbf{x}_{\mathcal{I}} + ({\mathbf{x}_{\mathrm{st}}}^*\mathbf{x}_{\mathcal{I}})^*) \\
&= \|\mathbf{x}_{\mathrm{st}}\|^2 + 2\varepsilon\mathrm{Re}({\mathbf{x}_{\mathrm{st}}}^*\mathbf{x}_{\mathcal{I}}).
\end{aligned}$$

Thus, $\widehat{\mathbf{x}}^*\widehat{\mathbf{x}} \in \mathbb{DR}$ and its standard part is equal to $\|\mathbf{x}_{\mathrm{st}}\|^2$. Since $\widehat{\mathbf{x}}$ is appreciable, $\mathbf{x}_{\mathrm{st}}$ is nonzero, and so $\|\mathbf{x}_{\mathrm{st}}\|^2 > 0$. □

Definition 3.18. [14] Let $\widehat{A} \in M_n(\mathbb{DC})$. We say that
(a) $\widehat{A}$ is *Hermitian* if $\widehat{A}^* = \widehat{A}$;
(b) $\widehat{A}$ is *skew-Hermitian* if $\widehat{A}^* = -\widehat{A}$;
(c) $\widehat{A}$ is *normal* if $\widehat{A}\widehat{A}^* = \widehat{A}^*\widehat{A}$; and
(d) $\widehat{A}$ is *unitary* if $\widehat{A}\widehat{A}^* = \widehat{A}^*\widehat{A} = I_n$.

Proposition 3.19. *Let $\widehat{A} \in M_n(\mathbb{DC})$. Then $\widehat{A}$ is*

(a) Hermitian if and only if A_{st} and $A_{\mathcal{I}}$ are Hermitian;

(b) skew-Hermitian if and only if A_{st} and $A_{\mathcal{I}}$ are skew-Hermitian;

(c) normal only if A_{st} is normal; and

(d) unitary if and only if A_{st} is unitary and ${A_{st}}^* A_{\mathcal{I}}$ is skew-Hermitian.

Proof. Let $\widehat{A} \in M_n(\mathbb{DC})$.

(a) Note that $\widehat{A}$ is Hermitian if and only if

$$\widehat{A}^* = (A_{\text{st}} + \varepsilon A_{\mathcal{I}})^* = {A_{\text{st}}}^* + \varepsilon {A_{\mathcal{I}}}^* = A_{\text{st}} + \varepsilon A_{\mathcal{I}} = \widehat{A}$$

if and only if ${A_{\text{st}}}^* = A_{\text{st}}$ and ${A_{\mathcal{I}}}^* = A_{\mathcal{I}}$ (i.e., A_{st} and $A_{\mathcal{I}}$ are Hermitian).

(b) Note that $\widehat{A}$ is skew-Hermitian if and only if

$$\widehat{A}^* = (A_{\text{st}} + \varepsilon A_{\mathcal{I}})^* = {A_{\text{st}}}^* + \varepsilon {A_{\mathcal{I}}}^* = -A_{\text{st}} - \varepsilon A_{\mathcal{I}} = -\widehat{A}$$

if and only if ${A_{\text{st}}}^* = -A_{\text{st}}$ and ${A_{\mathcal{I}}}^* = -A_{\mathcal{I}}$ (i.e., A_{st} and $A_{\mathcal{I}}$ are skew-Hermitian).

(c) Suppose $\widehat{A}$ is normal. Then,

$$\begin{aligned}
\widehat{A}\widehat{A}^* &= (A_{\text{st}} + \varepsilon A_{\mathcal{I}})(A_{\text{st}} + \varepsilon A_{\mathcal{I}})^* \\
&= (A_{\text{st}} + \varepsilon A_{\mathcal{I}})({A_{\text{st}}}^* + \varepsilon {A_{\mathcal{I}}}^*) \\
&= A_{\text{st}}{A_{\text{st}}}^* + \varepsilon(A_{\text{st}}{A_{\mathcal{I}}}^* + A_{\mathcal{I}}{A_{\text{st}}}^*) \\
&= {A_{\text{st}}}^* A_{\text{st}} + \varepsilon({A_{\text{st}}}^* A_{\mathcal{I}} + {A_{\mathcal{I}}}^* A_{\text{st}}) \\
&= ({A_{\text{st}}}^* + \varepsilon {A_{\mathcal{I}}}^*)(A_{\text{st}} + \varepsilon A_{\mathcal{I}}) \\
&= (A_{\text{st}} + \varepsilon A_{\mathcal{I}})^*(A_{\text{st}} + \varepsilon A_{\mathcal{I}}) \\
&= \widehat{A}^*\widehat{A},
\end{aligned}$$

and so $A_{\text{st}}{A_{\text{st}}}^* = {A_{\text{st}}}^* A_{\text{st}}$, i.e., A_{st} is normal.

(d). Note that $\widehat{A}$ is unitary if and only if

$$\begin{aligned}
\widehat{A}\widehat{A}^* &= A_{\text{st}}{A_{\text{st}}}^* + \varepsilon(A_{\text{st}}{A_{\mathcal{I}}}^* + A_{\mathcal{I}}{A_{\text{st}}}^*) \\
&= A_{\text{st}}{A_{\text{st}}}^* + \varepsilon(A_{\text{st}}{A_{\mathcal{I}}}^* + (A_{\text{st}}{A_{\mathcal{I}}}^*)^*) \\
&= I_n \\
&= {A_{\text{st}}}^* A_{\text{st}} + \varepsilon({A_{\text{st}}}^* A_{\mathcal{I}} + ({A_{\text{st}}}^* A_{\mathcal{I}})^*) \\
&= {A_{\text{st}}}^* A_{\text{st}} + \varepsilon({A_{\text{st}}}^* A_{\mathcal{I}} + {A_{\mathcal{I}}}^* A_{\text{st}}) \\
&= \widehat{A}^*\widehat{A}
\end{aligned}$$

if and only if $A_{\text{st}}{A_{\text{st}}}^* = I_n = {A_{\text{st}}}^* A_{\text{st}}$, $A_{\text{st}}{A_{\mathcal{I}}}^* = -(A_{\text{st}}{A_{\mathcal{I}}}^*)^*$, and ${A_{\text{st}}}^* A_{\mathcal{I}} = -({A_{\text{st}}}^* A_{\mathcal{I}})^*$, i.e., A_{st} is unitary and both $A_{\text{st}}{A_{\mathcal{I}}}^*$ and ${A_{\text{st}}}^* A_{\mathcal{I}}$ are skew-Hermitian.

We only need to show that $A_{\text{st}}A_{\mathcal{I}}{}^*$ is skew-Hermitian if and only if ${A_{\text{st}}}^*A_{\mathcal{I}}$ is skew-Hermitian. Indeed, since A_{st} is unitary, $A_{\text{st}}A_{\mathcal{I}}{}^*$ is skew-Hermitian if and only if

$$\begin{aligned} A_{\text{st}}A_{\mathcal{I}}{}^* &= -A_{\mathcal{I}}{A_{\text{st}}}^* \\ {A_{\text{st}}}^*(A_{\text{st}}A_{\mathcal{I}}{}^*)A_{\text{st}} &= -{A_{\text{st}}}^*(A_{\mathcal{I}}{A_{\text{st}}}^*)A_{\text{st}} \\ ({A_{\text{st}}}^*A_{\text{st}})A_{\mathcal{I}}{}^*A_{\text{st}} &= -{A_{\text{st}}}^*A_{\mathcal{I}}({A_{\text{st}}}^*A_{\text{st}}) \\ A_{\mathcal{I}}{}^*A_{\text{st}} &= -{A_{\text{st}}}^*A_{\mathcal{I}} \end{aligned}$$

if and only if $A_{\mathcal{I}}{}^*A_{\text{st}}$ is skew-Hermitian. □

Definition 3.20 (Similar Matrices over $\mathbb{DC}$)**.** [15] Let $\widehat{A} \in M_n(\mathbb{DC})$ and $\widehat{B} \in M_n(\mathbb{DC})$. Then, we say that $\widehat{A}$ is *similar* to $\widehat{B}$ (over $\mathbb{DC}$) if $\widehat{A} = \widehat{P}\widehat{B}\widehat{P}^{-1}$ (equivalently, $\widehat{B} = \widehat{P}^{-1}\widehat{A}\widehat{P}$) for some invertible $\widehat{P} \in M_n(\mathbb{DC})$. Moreover, if $\widehat{A}$ is similar to a diagonal matrix $\widehat{D} = D_{\text{st}} + \varepsilon D_{\mathcal{I}}$ (i.e., D_{st} and $D_{\mathcal{I}}$ are both diagonal matrices) then we say that $\widehat{A}$ is *diagonalizable*. If in addition, $\widehat{P}$ is unitary, then we say that $\widehat{A}$ is *unitarily diagonalizable* or $\widehat{A}$ is *unitarily similar* to a diagonal matrix $\widehat{D}$.

Lemma 3.21. *Let $D := [d_{ij}] \in M_n(\mathbb{C})$, and $F := [f_{ij}] \in M_n(\mathbb{C})$ such that D is a diagonal matrix with distinct diagonal entries (i.e., $d_{ii} \neq d_{jj}$ whenever $i \neq j$ for all $i, j \in \{1, \ldots, n\}$ and F has all zero diagonal entries (i.e., $f_{ii} = 0$ for all $i \in \{1, \ldots, n\}$. Then there exists $S := [s_{ij}] \in M_n(\mathbb{C})$ such that*

$$DS - SD = F.$$

Proof. Take $S := [s_{ij}] \in M_n(\mathbb{C})$ such that

$$s_{ij} = \begin{cases} 0 & \text{if } i = j \\ \frac{f_{ij}}{d_{ii} - d_{jj}} & \text{if } i \neq j \end{cases}.$$

Then

$$\begin{aligned} e_i^T(DS - SD)e_j &= e_i^T DSe_j - e_i^T SDe_j \\ &= d_{ii}s_{ij} - d_{jj}s_{ij} \\ &= (d_{ii} - d_{jj})\, s_{ij} \\ &= \begin{cases} 0 & \text{if } i = j \\ f_{ij} & \text{if } i \neq j \end{cases} \\ &= e_i^T Fe_j. \end{aligned}$$

Thus, $DS - SD = F$. □

Theorem 3.22. *Let $\widehat{A} \in M_n(\mathbb{DC})$. If A_{st} has n distinct eigenvalues, then $\widehat{A}$ is diagonalizable.*

Proof. Suppose A_{st} has n distinct eigenvalues. Then A_{st} is diagonalizable, i.e., there exists a diagonal matrix D such that $D = P^{-1}A_{\text{st}}P$ (equivalently, $A_{\text{st}} = PDP^{-1}$) for some invertible matrix $P \in M_n(\mathbb{C})$. Consider the equation

$$(P + \varepsilon Q)^{-1}(A_{\text{st}} + \varepsilon A_{\mathcal{I}})(P + \varepsilon Q) = D + \varepsilon C \tag{3.16}$$

where $C = \text{diag}(P^{-1}A_{\mathcal{I}}P)$, i.e., C is a diagonal matrix such that its main diagonal is identical to the main diagonal of $P^{-1}A_{\mathcal{I}}P$. We wish to solve for Q that will satisfy (3.16) to conclude that $\widehat{A}$ is diagonalizable. Note that (3.16) is equivalent to

$$P^{-1}A_{\text{st}}Q + P^{-1}A_{\mathcal{I}}P - P^{-1}QD = C \tag{3.17}$$

or

$$DP^{-1}Q - P^{-1}QD = C - P^{-1}A_{\mathcal{I}}P \tag{3.18}$$

which is of the form

$$DS - SD = F, \tag{3.19}$$

where $F := [f_{ij}] = C - P^{-1}A_{\mathcal{I}}P$ and $S = P^{-1}Q$. By our previous lemma, (3.19) has a solution S. Since $S = P^{-1}Q$, $Q = PS$ will give a solution to both (3.18) and (3.16). Hence, $\widehat{A}$ is diagonalizable. □

An important consequence of this theorem is a technique to find the eigenvalues of an $n \times n$ dual complex matrix, where its standard part has n distinct eigenvalues, as shown by the following theorem.

Corollary 3.23. *Let $\widehat{A} \in M_n(\mathbb{DC})$ such that A_{st} has n distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then for all $i \in \{1, 2, \ldots, n\}$, $\lambda_i + \mu_i\varepsilon$ is an eigenvalue of $\widehat{A}$, where $diag(\mu_1, \mu_2, \ldots, \mu_n) = diag(P^{-1}A_{\mathcal{I}}P)$ and $P^{-1}A_{st}P = D$ is a diagonal matrix for some invertible $P \in M_n(\mathbb{C})$.*

Proof. Since A_{st} has n distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, $\widehat{A}$ is diagonalizable by the preceding theorem, and so there exists an invertible $P := \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 & \ldots & \mathbf{x}_n \end{pmatrix} \in M_n(\mathbb{C})$ and $Q := \begin{pmatrix} \mathbf{y}_1 & \mathbf{y}_2 & \ldots & \mathbf{y}_n \end{pmatrix} \in M_n(\mathbb{C})$ such that $\mathbf{x}_i \in \mathbb{C}^n$ is an eigenvector of A_{st} corresponding to λ_i i.e.,

$$A_{\text{st}}\mathbf{x}_i = \lambda_i\mathbf{x}_i \tag{3.20}$$

for all $i \in \{1, 2, \ldots, n\}$ and

$$(P + \varepsilon Q)^{-1}(A_{\text{st}} + \varepsilon A_{\mathcal{I}})(P + \varepsilon Q) = D + \varepsilon C, \tag{3.21}$$

where $D = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) \in M_n(\mathbb{C})$ and

$$C := \text{diag}(\mu_1, \mu_2, \ldots, \mu_n) = \text{diag}(P^{-1} A_{\mathcal{I}} P)$$

are diagonal matrices. Note that (3.21) is further equivalent to (3.17). Premultiplying (3.17) by P, postmultiplying (3.17) by e_i, and rearranging yields

$$A_{\text{st}}\mathbf{y}_i + A_{\mathcal{I}}\mathbf{x}_i = \lambda_i \mathbf{y}_i + \mu_i \mathbf{x}_i. \tag{3.22}$$

Combining (3.20) and (3.22) and by recalling the fact that $\mathbf{0} \neq \mathbf{x}_i$ implies that $\lambda_i + \mu_i\varepsilon \in \mathbb{DC}$ is an eigenvalue of $\widehat{A}$ for all i. □

Proposition 3.24. *Let $\widehat{A} = A_{st} + \varepsilon A_{\mathcal{I}} \in M_n(\mathbb{DC})$ and $\widehat{B} = B_{st} + \varepsilon B_{\mathcal{I}} \in M_n(\mathbb{DC})$ be similar matrices such that $\widehat{A}$ has n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$. Then, $\widehat{A}$ and $\widehat{B}$ have the same set of eigenvalues, i.e., $\sigma(\widehat{A}) = \sigma(\widehat{B})$.*

Proof. Let $\lambda_i + \mu_i\varepsilon \in \sigma(\widehat{A})$ correspond to the (appreciable) eigenvector $\mathbf{x}_i + \varepsilon\mathbf{y}_i$ of $\widehat{A}$ for all $i \in \{1, \ldots, n\}$. Since $\widehat{A}$ and $\widehat{B}$ are similar, there exists an invertible $\widehat{P} = P_{\text{st}} + \varepsilon P_{\mathcal{I}} \in M_n(\mathbb{DC})$ such that $\widehat{A} = \widehat{P}^{-1}\widehat{B}\widehat{P}$. Postmultiplying both sides by $\mathbf{x} + \varepsilon\mathbf{y}$ and rearranging yields

$$(\lambda_i + \mu_i\varepsilon)(P_{\text{st}} + \varepsilon P_{\mathcal{I}})(\mathbf{x}_i + \varepsilon\mathbf{y}_i) = (B_{\text{st}} + \varepsilon B_{\mathcal{I}})(P_{\text{st}} + \varepsilon P_{\mathcal{I}})(\mathbf{x}_i + \varepsilon\mathbf{y}_i).$$

Now, we only need to show that $\mathbf{x}_i' + \varepsilon\mathbf{y}_i' := (P_{\text{st}} + \varepsilon P_{\mathcal{I}})(\mathbf{x}_i + \varepsilon\mathbf{y}_i)$ is appreciable, i.e., $P_{\text{st}}\mathbf{x}_i$ is nonzero so that $\mathbf{x}_i' + \varepsilon\mathbf{y}_i'$ is an eigenvector of $\widehat{B}$ corresponding to $\lambda_i + \mu_i\varepsilon$ for all $i \in \{1, \ldots, n\}$. Indeed, recall that $\widehat{P}$ is invertible if and only if P_{st} is invertible. Thus, if $P_{\text{st}}\mathbf{x}_i = \mathbf{0}$, then $\mathbf{x}_i = \mathbf{0}$, and so $\mathbf{x}_i + \varepsilon\mathbf{y}_i$ is not appreciable, a contradiction. Hence, $\lambda_i + \mu_i\varepsilon \in \sigma(\widehat{B})$ for all $i \in \{1, \ldots, n\}$, i.e., $\sigma(\widehat{A}) \subseteq \sigma(\widehat{B})$. We can apply the same strategy to show the reverse inclusion $\sigma(\widehat{A}) \supseteq \sigma(\widehat{B})$. Therefore, $\sigma(\widehat{A}) = \sigma(\widehat{B})$. □

Definition 3.25 (Householder Matrices over $\mathbb{DC}$)**.** [7] Let $\widehat{\mathbf{x}} \in \mathbb{DC}^n$ be appreciable. The *Householder matrix* $\widehat{Q}_{\widehat{\mathbf{x}}} \in M_n(\mathbb{DC})$ associated with $\widehat{\mathbf{x}}$ is defined as

$$\widehat{Q}_{\widehat{\mathbf{x}}} = I_n - \frac{2}{\widehat{\mathbf{x}}^*\widehat{\mathbf{x}}}\widehat{\mathbf{x}}\widehat{\mathbf{x}}^*,$$

where $\widehat{\mathbf{x}}^*\widehat{\mathbf{x}} = \|\mathbf{x}_{\text{st}}\|^2 + 2\varepsilon\text{Re}(\mathbf{x}_{\text{st}}{}^*\mathbf{x}_{\mathcal{I}})$.

Theorem 3.26 (Properties of Householder Matrices over $\mathbb{DC}$)**.** *[7] Let $\widehat{Q}_{\widehat{\boldsymbol{x}}} \in M_n(\mathbb{DC})$ be the Householder matrix associated with $\widehat{\boldsymbol{x}}$ such that $\widehat{\boldsymbol{x}} \in \mathbb{DC}^n$ is appreciable. Then, $\widehat{Q}_{\widehat{\boldsymbol{x}}}$ is Hermitian and unitary.*

Proof. Since $\widehat{\mathbf{x}}$ is appreciable, $\widehat{\mathbf{x}}^*\widehat{\mathbf{x}}$ is also appreciable. Note that for any appreciable $\widehat{a} \in \mathbb{DC}$, $\left(\frac{1}{\widehat{a}}\right)^* = \frac{1}{\widehat{a}^*}$. Thus,

$$\widehat{Q}_{\widehat{\mathbf{x}}}{}^* = \left(I_n - \frac{2}{\widehat{\mathbf{x}}^*\widehat{\mathbf{x}}}\widehat{\mathbf{x}}\widehat{\mathbf{x}}^*\right)^* = I_n{}^* - \frac{2}{(\widehat{\mathbf{x}}^*\widehat{\mathbf{x}})^*}(\widehat{\mathbf{x}}\widehat{\mathbf{x}}^*)^* = I_n - \frac{2}{\widehat{\mathbf{x}}^*\widehat{\mathbf{x}}}\widehat{\mathbf{x}}\widehat{\mathbf{x}}^* = \widehat{Q}_{\widehat{\mathbf{x}}}.$$

In particular,

$$\begin{aligned}
\widehat{Q}_{\widehat{\mathbf{x}}}{}^*\widehat{Q}_{\widehat{\mathbf{x}}} &= \widehat{Q}_{\widehat{\mathbf{x}}}^2 \\
&= \left(I_n - \frac{2}{\widehat{\mathbf{x}}^*\widehat{\mathbf{x}}}\widehat{\mathbf{x}}\widehat{\mathbf{x}}^*\right)^2 \\
&= I_n - \frac{4}{\widehat{\mathbf{x}}^*\widehat{\mathbf{x}}}\widehat{\mathbf{x}}\widehat{\mathbf{x}}^* + \frac{4}{(\widehat{\mathbf{x}}^*\widehat{\mathbf{x}})^2}(\widehat{\mathbf{x}}(\widehat{\mathbf{x}}^*\widehat{\mathbf{x}})\widehat{\mathbf{x}}^*) \\
&= I_n - \frac{4}{\widehat{\mathbf{x}}^*\widehat{\mathbf{x}}}\widehat{\mathbf{x}}\widehat{\mathbf{x}}^* + \frac{4}{\widehat{\mathbf{x}}^*\widehat{\mathbf{x}}}\widehat{\mathbf{x}}\widehat{\mathbf{x}}^* \\
&= I_n \\
&= \widehat{Q}_{\widehat{\mathbf{x}}}\widehat{Q}_{\widehat{\mathbf{x}}}{}^*.
\end{aligned}$$

□

Theorem 3.27 (Schur's Triangularization Theorem for $M_n(\mathbb{DC})$)**.** *Let $\widehat{A} \in M_n(\mathbb{DC})$ such that A_{st} has n distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then, $\widehat{A}$ is unitarily similar to an upper triangular matrix $\widehat{T}_n \in M_n(\mathbb{DC})$, i.e., there exists unitary $\widehat{U}$ such that $\widehat{U}^*\widehat{A}\widehat{U} = \widehat{T}_n$ is upper triangular and the diagonal entries of $\widehat{T}_n$ are eigenvalues of $\widehat{A}$ and are of the form $\lambda_1 + \varepsilon\mu_1, \lambda_2 + \varepsilon\mu_2, \ldots, \lambda_n + \varepsilon\mu_n$ for some $\mu_1, \ldots, \mu_n \in \mathbb{C}$.*

Proof. We will proceed by induction on n.
Base Step. The claim is clear for $n = 1$.
Inductive Step. Suppose the claim holds for $n-1 \times n-1$ dual complex matrices with $n-1$ distinct eigenvalues ($n \geq 2$). Since A_{st} has n distinct eigenvalues, $\widehat{A}$ has a nonempty spectrum. Thus, we can let $\widehat{\mathbf{v}} = \begin{pmatrix}\widehat{v}_1 & \widehat{v}_2 & \ldots & \widehat{v}_i & \ldots & \widehat{v}_n\end{pmatrix}^T \in \mathbb{DC}^n$ be an eigenvector associated with the eigenvalue $\lambda_1 + \mu_1\varepsilon$ of $\widehat{A}$, where $\widehat{v}_i$ is appreciable for some $i \in \{1, 2, \ldots, n\}$ and let $\widehat{\mathbf{y}} = \widehat{v}_i{}^*\widehat{\mathbf{v}} := \begin{pmatrix}\widehat{y}_1 & \widehat{y}_2 & \ldots & \widehat{y}_i & \ldots & \widehat{y}_n\end{pmatrix}^T \in \mathbb{DC}^n$. By Theorem 3.17.6, $\widehat{y}_i = \widehat{v}_i{}^*\widehat{v}_i \in \mathbb{DR}$ and its standard part is strictly positive. Now, let $\widehat{r} = \frac{\|\mathbf{y}_{\text{st}}\|}{\|\mathbf{y}_{\text{st}}\|^2 + \varepsilon \text{Re}(\mathbf{y}_{\text{st}}{}^*\mathbf{y}_{\mathcal{I}})} \in \mathbb{DR}$ and define

$$\widehat{\mathbf{x}} := \begin{pmatrix}\widehat{x}_1 & \widehat{x}_2 & \ldots & \widehat{x}_i & \ldots & \widehat{x}_n\end{pmatrix}^T = \begin{cases} -\widehat{r}\widehat{\mathbf{y}} & \text{if } r_{\text{st}} y_{i,\text{st}} = 1 \\ \widehat{r}\widehat{\mathbf{y}} & \text{otherwise.} \end{cases}$$

Thus, $\widehat{\mathbf{x}}^*\widehat{\mathbf{x}} = 1$, where $\widehat{x}_i = \pm\widehat{r}\widehat{y}_i \in \mathbb{DR}$ and $x_{i,\mathrm{st}} \neq 1$. In particular, ${\widehat{x}_i}^* = \widehat{x}_i$. Let $\widehat{\mathbf{u}} = \begin{pmatrix} \widehat{u}_1 & \widehat{u}_2 & \dots & \widehat{u}_i & \dots & \widehat{u}_n \end{pmatrix}^T = \widehat{\mathbf{x}} - e_i$. Then,

$$\widehat{\mathbf{u}}^*\widehat{\mathbf{u}} = (\widehat{\mathbf{x}} - e_i)^*(\widehat{\mathbf{x}} - e_i) = \widehat{\mathbf{x}}^*\widehat{\mathbf{x}} - \widehat{\mathbf{x}}^*e_i - {e_i}^*\widehat{\mathbf{x}} + {e_i}^*e_i = 2 - 2\widehat{x}_i$$

and

$$\widehat{\mathbf{u}}^*\widehat{\mathbf{x}} = (\widehat{\mathbf{x}} - e_i)^*\widehat{\mathbf{x}} = \widehat{\mathbf{x}}^*\widehat{\mathbf{x}} - {e_i}^*\widehat{\mathbf{x}} = 1 - \widehat{x}_i.$$

Notice that $\widehat{\mathbf{u}}$ is appreciable since $x_{i,\mathrm{st}} \neq 1$. Thus, $\widehat{Q}_{\widehat{\mathbf{u}}}$ exists and

$$\widehat{Q}_{\widehat{\mathbf{u}}}\widehat{\mathbf{x}} = \widehat{\mathbf{x}} - \frac{2}{\widehat{\mathbf{u}}^*\widehat{\mathbf{u}}}\widehat{\mathbf{u}}\widehat{\mathbf{u}}^*\widehat{\mathbf{x}} = \widehat{\mathbf{x}} - \frac{2}{2 - 2\widehat{x}_i}\widehat{\mathbf{u}}(1 - \widehat{x}_i) = \widehat{\mathbf{x}} - \widehat{\mathbf{u}} = e_i.$$

Now, let $P \in M_n(\mathbb{R})$ be a type III elementary matrix such that $P\widehat{Q}_{\widehat{\mathbf{u}}}\widehat{\mathbf{x}} = e_1$. Note that $P^{-1} = P = P^*$ and P is a permutation matrix (hence unitary). Since $\widehat{Q}_{\widehat{\mathbf{u}}}$ is Hermitian and unitary, $\widehat{\mathbf{x}} = \widehat{Q}_{\widehat{\mathbf{u}}}Pe_1$ and $\widehat{\mathbf{x}}^* = e_1^T P\widehat{Q}_{\widehat{\mathbf{u}}}$. Thus, we can choose $\widehat{Q} = Q_{\mathrm{st}} + \varepsilon Q_{\mathcal{I}} = \widehat{Q}_{\widehat{\mathbf{u}}}P$ so that $\widehat{Q} = \begin{pmatrix} \widehat{\mathbf{x}} & \widehat{V} \end{pmatrix}$ and $\begin{pmatrix} \widehat{\mathbf{x}}^* \\ \widehat{V}^* \end{pmatrix} = \widehat{Q}^*$, where $\widehat{V} \in M_{n,n-1}(\mathbb{DC})$. Note that $\widehat{Q}^*\widehat{Q} = (\widehat{Q}_{\widehat{\mathbf{u}}}P)^*(\widehat{Q}_{\widehat{\mathbf{u}}}P) = P^*\widehat{Q}_{\widehat{\mathbf{u}}}^*Q_{\widehat{\mathbf{u}}}P = I_n$, which implies that $\widehat{Q}$ is unitary. Then

$$\begin{aligned}
\widehat{Q}^*\widehat{A}\widehat{Q} &= \widehat{Q}^*\widehat{A}\begin{pmatrix} \widehat{\mathbf{x}} & \widehat{V} \end{pmatrix} \\
&= \begin{pmatrix} \widehat{Q}^*\widehat{A}\widehat{\mathbf{x}} & \widehat{Q}\widehat{A}\widehat{V} \end{pmatrix} \\
&= \begin{pmatrix} \widehat{Q}^*(\lambda_1 + \mu_1\varepsilon)\widehat{\mathbf{x}} & \begin{pmatrix} \widehat{\mathbf{x}}^* \\ \widehat{V}^* \end{pmatrix}\widehat{A}\widehat{V} \end{pmatrix} \\
&= \begin{pmatrix} (\lambda_1 + \mu_1\varepsilon)e_1 & \begin{pmatrix} \widehat{\mathbf{x}}^*\widehat{A}\widehat{V} \\ \widehat{V}^*\widehat{A}\widehat{V} \end{pmatrix} \end{pmatrix} \\
&= \begin{pmatrix} \lambda_1 + \mu_1\varepsilon & \widehat{\mathbf{x}}^*\widehat{A}\widehat{V} \\ 0_{n-1,1} & \widehat{V}^*\widehat{A}\widehat{V} \end{pmatrix}.
\end{aligned}$$

Now, we need to show that ${V_{\mathrm{st}}}^*A_{\mathrm{st}}V_{\mathrm{st}}$ has $n-1$ distinct eigenvalues to apply the induction hypothesis. Indeed, note that

$${Q_{\mathrm{st}}}^*A_{\mathrm{st}}Q_{\mathrm{st}} = \begin{pmatrix} \lambda_1 & {\mathbf{x}_{\mathrm{st}}}^*A_{\mathrm{st}}V_{\mathrm{st}} \\ 0_{n-1,1} & {V_{\mathrm{st}}}^*A_{\mathrm{st}}V_{\mathrm{st}} \end{pmatrix}$$

is the standard part of $\widehat{Q}^*\widehat{A}\widehat{Q}$. Since $\widehat{Q}$ is unitary, $\widehat{Q}_{\mathrm{st}}$ is also unitary (by Proposition

3.19). Thus,

$$\begin{aligned}
0 &= \det\left(A_{\text{st}} - \lambda_k I_n\right) \\
&= \det\left({Q_{\text{st}}}^*\right)\det\left(A_{\text{st}} - \lambda_k I_n\right)\det\left(Q_{\text{st}}\right) \\
&= \det\left({Q_{\text{st}}}^*\left(A_{\text{st}} - \lambda_k I_n\right)Q_{\text{st}}\right) \\
&= \det\left({Q_{\text{st}}}^* A_{\text{st}} Q_{\text{st}} - \lambda_k I_n\right) \\
&= \det\begin{pmatrix} \lambda_1 - \lambda_k & {\mathbf{x}_{\text{st}}}^* A_{\text{st}} V_{\text{st}} \\ 0_{n-1,1} & {V_{\text{st}}}^* A_{\text{st}} V_{\text{st}} - \lambda_k I_{n-1} \end{pmatrix} \\
&= (\lambda_1 - \lambda_k)\det\left({V_{\text{st}}}^* A_{\text{st}} V_{\text{st}} - \lambda_k I_{n-1}\right)
\end{aligned}$$

for all $k \in \{2, 3, \ldots, n\}$. Since $\lambda_1, \ldots, \lambda_n$ are distinct, $\det\left({V_{\text{st}}}^* A_{\text{st}} V_{\text{st}} - \lambda_k I_{n-1}\right) = 0$, and so λ_k is an eigenvalue of ${V_{\text{st}}}^* A_{\text{st}} V_{\text{st}}$ for all $k \in \{2, 3, \ldots, n\}$ (which are distinct from one another). By induction hypothesis, there is a unitary $\widehat{R} \in M_{n-1}(\mathbb{DC})$ such that $\widehat{R}^*\widehat{V}^*\widehat{A}\widehat{V}\widehat{R} = T_{n-1} = [t_{ij}]$ is upper triangular, where $t_{i_k i_k} = \lambda_k + \mu_k\varepsilon$ is an eigenvalue of $\widehat{V}^*\widehat{A}\widehat{V}$ for all $k \in \{2, 3, \ldots n\}$. Now, let $\widehat{U} = \widehat{Q}\begin{pmatrix} 1 & 0_{1,n-1} \\ 0_{n-1,1} & \widehat{R} \end{pmatrix}$. Then,

$$\begin{aligned}
\widehat{U}^*\widehat{U} &= \left(\widehat{Q}\begin{pmatrix} 1 & 0_{1,n-1} \\ 0_{n-1,1} & \widehat{R} \end{pmatrix}\right)^*\widehat{Q}\begin{pmatrix} 1 & 0_{1,n-1} \\ 0_{n-1,1} & \widehat{R} \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0_{1,n-1} \\ 0_{n-1,1} & \widehat{R}^* \end{pmatrix}\widehat{Q}^*\widehat{Q}\begin{pmatrix} 1 & 0_{1,n-1} \\ 0_{n-1,1} & \widehat{R} \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0_{1,n-1} \\ 0_{n-1,1} & \widehat{R}^*\widehat{R} \end{pmatrix} \\
&= I_n.
\end{aligned}$$

In a similar fashion, we can show that $\widehat{U}\widehat{U}^* = I_n$. Thus, $\widehat{U}$ is unitary. In particular,

$$\begin{aligned}
\widehat{U}^*\widehat{A}\widehat{U} &= \begin{pmatrix} 1 & 0_{1,n-1} \\ 0_{n-1,1} & \widehat{R}^* \end{pmatrix}\widehat{Q}^*\widehat{A}\widehat{Q}\begin{pmatrix} 1 & 0_{1,n-1} \\ 0_{n-1,1} & \widehat{R} \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0_{1,n-1} \\ 0_{n-1,1} & \widehat{R}^* \end{pmatrix}\begin{pmatrix} \lambda_1 + \varepsilon\mu_1 & \widehat{\mathbf{x}}^*\widehat{A}\widehat{V} \\ 0_{n-1,1} & \widehat{V}^*\widehat{A}\widehat{V} \end{pmatrix}\begin{pmatrix} 1 & 0_{1,n-1} \\ 0_{n-1,1} & \widehat{R} \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0_{1,n-1} \\ 0_{n-1,1} & \widehat{R}^* \end{pmatrix}\begin{pmatrix} \lambda_1 + \varepsilon\mu_1 & \widehat{\mathbf{x}}^*\widehat{A}\widehat{V}\widehat{R} \\ 0_{n-1,1} & \widehat{V}^*\widehat{A}\widehat{V}\widehat{R} \end{pmatrix} \\
&= \begin{pmatrix} \lambda_1 + \varepsilon\mu_1 & \widehat{\mathbf{x}}^*\widehat{A}\widehat{V}\widehat{R} \\ 0_{n-1,1} & \widehat{R}^*\widehat{V}^*\widehat{A}\widehat{V}\widehat{R} \end{pmatrix}
\end{aligned}$$

$$= \begin{pmatrix} \lambda_1 + \varepsilon\mu_1 & \widehat{\mathbf{x}}^* \widehat{A}\widehat{V}\widehat{R} \\ 0_{n-1,1} & T_{n-1} \end{pmatrix}$$

which is upper triangular. Now, we only need to show that $\lambda_k + \varepsilon\mu_k \in \sigma(\widehat{A})$ for all $k \in \{2, 3, ..., n\}$. Indeed, we know that $\widehat{V}^* \widehat{A}\widehat{V}\widehat{\mathbf{w}}_k = (\lambda_k + \varepsilon\mu_k)\widehat{\mathbf{w}}_k$ for some eigenvector $\widehat{\mathbf{w}}_k \in \mathbb{DC}^{n-1}$. Let $\widehat{b}_k = \frac{\widehat{\mathbf{x}}^* \widehat{A}\widehat{V}\widehat{\mathbf{w}}_k}{\lambda_k + \varepsilon\mu_k - (\lambda_1 + \varepsilon\mu_1)} \in \mathbb{DC}$. Notice that

$$\widehat{A}\widehat{Q} \begin{pmatrix} \widehat{b}_k \\ \widehat{\mathbf{w}}_k \end{pmatrix} = \widehat{Q} \begin{pmatrix} \lambda_1 + \varepsilon\mu_1 & \widehat{\mathbf{x}}^* \widehat{A}\widehat{V} \\ 0_{n-1,1} & \widehat{V}^* \widehat{A}\widehat{V} \end{pmatrix} \begin{pmatrix} \widehat{b}_k \\ \widehat{\mathbf{w}}_k \end{pmatrix} = (\lambda_k + \varepsilon\mu_k)\widehat{Q} \begin{pmatrix} \widehat{b}_k \\ \widehat{\mathbf{w}}_k \end{pmatrix}.$$

Since $\widehat{\mathbf{w}}_k$ and $\widehat{Q}$ are appreciable, $\widehat{Q} \begin{pmatrix} \widehat{b}_k \\ \widehat{\mathbf{w}}_k \end{pmatrix}$ is appreciable. Hence, $\lambda_k + \varepsilon\mu_k$ is an eigenvalue of $\widehat{A}$ for all $k \in \{2, 3, \ldots n\}$. □

Chapter 4

Some Consequences of STT and Sylvester's Theorem for Dual Complex Matrices

Schur's triangularization theorem has a wide set of consequences. Some of these are the trace and determinant formulas of a matrix, Sylvester's theorem, and the Spectral Theorem for normal matrices (consequently, for Hermitian matrices as well). Since Schur's triangularization theorem has been proved, we take a look at some of its consequences.

4.1 Trace of a Dual Complex Matrix

Recall that the *trace* of $A = [a_{ij}] \in M_{m,n}(R)$, where R is a ring, is the sum of its main diagonal entries: $\text{tr } A = \sum_{i=1}^{q} a_{ii}$, where $q = \min\{m, n\}$ [7]. Thus for all $\widehat{A} := [\alpha_{ij}] + \varepsilon[\beta_{ij}] \in M_{m,n}(\mathbb{D}R)$,

$$\text{tr } \widehat{A} = \sum_{i=1}^{q} \alpha_{ii} + \varepsilon \sum_{i=1}^{q} \beta_{ii} = \text{tr}(A_{\text{st}}) + \varepsilon \text{tr}(A_{\mathcal{I}}).$$

Proposition 4.1. *Let $A = [a_{ij}] \in M_n(R)$ and $B = [b_{ij}] \in M_n(R)$, where R is a commutative ring. Then tr(AB) = tr(BA).*

Proof. Since R is a commutative ring,

$$\text{tr}(AB) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} b_{ji} = \sum_{j=1}^{n} \sum_{i=1}^{n} b_{ji} a_{ij} = \text{tr}(BA).$$

□

Theorem 4.2. *Let $\widehat{A} \in M_n(\mathbb{DC})$ such that A_{st} has n distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then*

$$tr\ \widehat{A} = \sum_{i=1}^{n} (\lambda_i + \varepsilon \mu_i),$$

where $\lambda_1 + \varepsilon\mu_1, \lambda_2 + \varepsilon\mu_2, \ldots, \lambda_n + \varepsilon\mu_n$ are eigenvalues of $\widehat{A}$.

Proof. Since A_{st} has n distinct eigenvalues, there exists a unitary $\widehat{U}$ and an upper triangular $\widehat{T}$ such that $\widehat{A} = \widehat{U}\widehat{T}\widehat{U}^*$ and $\operatorname{diag}(\widehat{T}) = \operatorname{diag}(\lambda_1 + \varepsilon\mu_1, \lambda_2 + \varepsilon\mu_2, \ldots, \lambda_n + \varepsilon\mu_n)$. Then

$$\operatorname{tr}(\widehat{A}) = \operatorname{tr}(\widehat{U}\widehat{T}\widehat{U}^*) = \operatorname{tr}(\widehat{U}^*\widehat{U}\widehat{T}) = \operatorname{tr}(\widehat{T}) = \sum_{i=1}^{n}(\lambda_i + \varepsilon\mu_i).$$

□

4.2 Determinant of a Dual Complex Matrix

Definition 4.3. [7] Let $\widehat{A} := [\alpha_{ij}] + \varepsilon[\beta_{ij}] \in M_n(\mathbb{DF})$. Then we define the determinant of $\widehat{A}$ similar to how determinant is defined over an arbitrary field $\mathbb{F}$, i.e.,

$$\det \widehat{A} = \begin{cases} \alpha_{11} + \varepsilon\beta_{11} & \text{if } n = 1 \\ \sum_{\sigma \in S_n} \operatorname{sgn}\sigma \prod_{i=1}^{n}(\alpha_{i\sigma(i)} + \varepsilon\beta_{i\sigma(i)}) & \text{if } n > 1, \end{cases}$$

where S_n is the symmetric group on n letters.

Proposition 4.4. *Let* $\widehat{A} := [\alpha_{ij}] + \varepsilon[\beta_{ij}] \in M_n(\mathbb{DF})$. *Then*

$$\prod_{i=1}^{\eta}(\alpha_{i\sigma(i)} + \varepsilon\beta_{i\sigma(i)}) = \prod_{i=1}^{\eta}\alpha_{i\sigma(i)} + \varepsilon\sum_{k=1}^{\eta}\prod_{\substack{j=1 \\ j\neq k}}^{\eta}\alpha_{j\sigma(j)}\beta_{k\sigma(k)}$$

for all $\eta \in \mathbb{N}, \eta > 1$.

Proof. Suppose $\eta = 2$. Then

$$\begin{aligned} \prod_{i=1}^{\eta}(\alpha_{i\sigma(i)} + \varepsilon\beta_{i\sigma(i)}) &= (\alpha_{1\sigma(1)} + \varepsilon\beta_{1\sigma(1)})(\alpha_{2\sigma(2)} + \varepsilon\beta_{2\sigma(2)}) \\ &= \alpha_{1\sigma(1)}\alpha_{2\sigma(2)} + \varepsilon(\alpha_{1\sigma(1)}\beta_{2\sigma(2)} + \beta_{1\sigma(1)}\alpha_{2\sigma(2)}) \\ &= \prod_{i=1}^{\eta}\alpha_{i\sigma(i)} + \varepsilon\sum_{k=1}^{\eta}\prod_{\substack{j=1 \\ j\neq k}}^{\eta}\alpha_{j\sigma(j)}\beta_{k\sigma(k)}. \end{aligned}$$

Now, assume that the claim above holds for $\eta = \kappa$, where $\kappa \in \mathbb{N}$ and $\kappa > 1$. Then,

$$\begin{aligned}
\prod_{i=1}^{\kappa+1}(\alpha_{i\sigma(i)} + \varepsilon\beta_{i\sigma(i)}) &= \left(\prod_{i=1}^{\kappa}\alpha_{i\sigma(i)} + \varepsilon\sum_{k=1}^{\kappa}\prod_{\substack{j=1\\j\neq k}}^{\kappa}\alpha_{j\sigma(j)}\beta_{k\sigma(k)}\right)\left(\alpha_{(\kappa+1)\sigma(\kappa+1)} + \varepsilon\beta_{(\kappa+1)\sigma(\kappa+1)}\right) \\
&= \left(\prod_{i=1}^{\kappa}\alpha_{i\sigma(i)}\right)\left(\alpha_{(\kappa+1)\sigma(\kappa+1)}\right) \\
&\quad + \varepsilon\left(\left(\prod_{i=1}^{\kappa}\alpha_{i\sigma(i)}\right)\beta_{(\kappa+1)\sigma(\kappa+1)} + \left(\sum_{k=1}^{\kappa}\prod_{\substack{j=1\\j\neq k}}^{\kappa}\alpha_{j\sigma(j)}\beta_{k\sigma(k)}\right)\alpha_{(\kappa+1)\sigma(\kappa+1)}\right) \\
&= \prod_{i=1}^{\kappa+1}\alpha_{i\sigma(i)} + \varepsilon\left(\left(\prod_{i=1}^{\kappa}\alpha_{i\sigma(i)}\right)\beta_{(\kappa+1)\sigma(\kappa+1)} + \sum_{k=1}^{\kappa}\prod_{\substack{j=1\\j\neq k}}^{\kappa+1}\alpha_{j\sigma(j)}\beta_{k\sigma(k)}\right) \\
&= \prod_{i=1}^{\kappa+1}\alpha_{i\sigma(i)} + \varepsilon\left(\sum_{k=1}^{\kappa+1}\prod_{\substack{j=1\\j\neq k}}^{\kappa+1}\alpha_{j\sigma(j)}\beta_{k\sigma(k)}\right).
\end{aligned}$$

Thus, the claim holds for $\eta = \kappa + 1$. By the principle of mathematical induction,

$$\prod_{i=1}^{\eta}(\alpha_{i\sigma(i)} + \varepsilon\beta_{i\sigma(i)}) = \prod_{i=1}^{\eta}\alpha_{i\sigma(i)} + \varepsilon\sum_{k=1}^{\eta}\prod_{\substack{j=1\\j\neq k}}^{\eta}\alpha_{j\sigma(j)}\beta_{k\sigma(k)}$$

for all $\eta \in \mathbb{N}, \eta > 1$. □

Definition 4.5. [7] Let $A := [a_{ij}] \in M_n(\mathbb{F})$, where $n \geq 2$. Then, we define the *adjugate* of A (denoted as adj A) as the transposed matrix of cofactors of A, i.e.,

$$\text{adj } A = [(-1)^{i+j}\det A[\{j\}^c, \{i\}^c]],$$

where $A[\{j\}^c, \{i\}^c]]$ is the matrix formed by deleting the jth row and i^{th} column of A. Moreover, if $n = 1$, we define adj $A := 1$.

Lemma 4.6. *Let $A := [a_{ij}] \in M_n(\mathbb{F})$, $B := [b_{ij}] \in M_n(\mathbb{F})$ and for all $k = 1, 2, \ldots, n$, define*

$$F_k(A, B) = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{k-1,1} & a_{k-1,2} & \ldots & a_{k-1,n} \\ b_{k1} & b_{k2} & \ldots & b_{kn} \\ a_{k+1,1} & a_{k+1,2} & \ldots & a_{k+1,n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{pmatrix},$$

i.e., $F_k(A,B)$ is the matrix formed by replacing the kth row of A with the kth row of B. Then,

$$\sum_{k=1}^{n} det\ F_k(A,B) = tr(B(adj\ A)).$$

Proof. For all $k \in \{1,\dots,n\}$, we will perform cofactor expansion on the kth row of $F_k(A,B)$ to compute det $F_k(A,B)$, i.e.,

$$\det\ F_k(A,B) = \sum_{i=1}^{n} b_{ki}C_{ki} = \underbrace{\begin{pmatrix} b_{k1} & \dots & b_{kn} \end{pmatrix}}_{k^{\text{th}} \text{ row of } B} \quad \underbrace{\begin{pmatrix} C_{k1} \\ \vdots \\ C_{kn} \end{pmatrix}}_{k^{\text{th}} \text{ column of adj } A},$$

where $C_{ij} = (-1)^{i+j}\det\ A[\{i\}^c\ \{j\}^c]$ is the $(i,j)^{\text{th}}$ cofactor of A. In particular,

$$\sum_{k=1}^{n} \det\ F_k(A,B) = \text{tr}(B(\text{adj } A)).$$

□

Theorem 4.7. *Let $\widehat{A} := [\alpha_{ij}] + \varepsilon[\beta_{ij}] \in M_n(\mathbb{DF})$. Then*

$$det\ \widehat{A} = det\ A_{st} + \varepsilon tr(A_{\mathcal{I}}(adj\ A_{st})).$$

for all $n > 1$ (similar to [13]).

Proof. From the definition and from the previous lemma,

$$\begin{aligned}
\det\ \widehat{A} &= \sum_{\sigma\in S_n} \text{sgn}\sigma \prod_{i=1}^{n} (\alpha_{i\sigma(i)} + \varepsilon\beta_{i\sigma(i)}) \\
&= \sum_{\sigma\in S_n} \text{sgn}\sigma \left(\prod_{i=1}^{n} \alpha_{i\sigma(i)} + \varepsilon \sum_{k=1}^{n} \prod_{\substack{j=1 \\ j\neq k}}^{n} \alpha_{j\sigma(j)}\beta_{k\sigma(k)} \right) \\
&= \sum_{\sigma\in S_n} \text{sgn}\sigma \prod_{i=1}^{n} \alpha_{i\sigma(i)} + \varepsilon \sum_{\sigma\in S_n} \text{sgn}\sigma \sum_{k=1}^{n} \prod_{\substack{j=1 \\ j\neq k}}^{n} \alpha_{j\sigma(j)}\beta_{k\sigma(k)} \\
&= \det A_{\text{st}} + \varepsilon \sum_{k=1}^{n} \left(\sum_{\sigma\in S_n} \text{sgn}\sigma \prod_{\substack{j=1 \\ j\neq k}}^{n} \alpha_{j\sigma(j)}\beta_{k\sigma(k)} \right) \\
&= \det A_{\text{st}} + \varepsilon \sum_{k=1}^{n} \det\ F_k(A_{\text{st}}, A_{\mathcal{I}}) \\
&= \det A_{\text{st}} + \varepsilon \text{tr}(A_{\mathcal{I}}(\text{adj} A_{\text{st}})).
\end{aligned}$$

□

The next proposition will help us recall some of the basic properties of the determinant, trace, and adjugate of a matrix $A \in M_n(\mathbb{F})$.

Proposition 4.8. *[7] Let $A \in M_n(\mathbb{F})$. Then*

(a). adj $(AB) = (adj\ B)(adj\ A)$; and

(b). $(adj\ A)A = A(adj\ A) = (det\ A)I_n$

Proposition 4.9 (Multiplicativity of the Determinant)**.** *Let $\widehat{A} = A_{st} + \varepsilon A_{\mathcal{I}} \in M_n(\mathbb{DF})$ and $\widehat{B} = B_{st} + \varepsilon B_{\mathcal{I}} \in M_n(\mathbb{DF})$. Then, $det(\widehat{A}\widehat{B}) = det\widehat{A}\ det\widehat{B}$.*

Proof. By computation, we have

$$\begin{aligned}
\det(\widehat{A}\widehat{B}) &= \det((A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}})(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}})) \\
&= \det(A_{\mathrm{st}}B_{\mathrm{st}} + \varepsilon(A_{\mathrm{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\mathrm{st}})) \\
&= \det(A_{\mathrm{st}}B_{\mathrm{st}}) + \varepsilon\mathrm{tr}((A_{\mathrm{st}}B_{\mathcal{I}} + A_{\mathcal{I}}B_{\mathrm{st}})\mathrm{adj}(A_{\mathrm{st}}B_{\mathrm{st}})) \\
&= \det A_{\mathrm{st}}\ \det B_{\mathrm{st}} + \varepsilon\mathrm{tr}(A_{\mathrm{st}}B_{\mathcal{I}}\mathrm{adj}(A_{\mathrm{st}}B_{\mathrm{st}}) + A_{\mathcal{I}}B_{\mathrm{st}}\mathrm{adj}(A_{\mathrm{st}}B_{\mathrm{st}})) \\
&= \det A_{\mathrm{st}}\ \det B_{\mathrm{st}} + \varepsilon(\mathrm{tr}(A_{\mathrm{st}}B_{\mathcal{I}}\mathrm{adj}(A_{\mathrm{st}}B_{\mathrm{st}})) + \mathrm{tr}(A_{\mathcal{I}}B_{\mathrm{st}}\mathrm{adj}(A_{\mathrm{st}}B_{\mathrm{st}}))) \\
&= \det A_{\mathrm{st}}\ \det B_{\mathrm{st}} + \varepsilon(\mathrm{tr}(A_{\mathrm{st}}B_{\mathcal{I}}(\mathrm{adj}B_{\mathrm{st}})(\mathrm{adj}A_{\mathrm{st}})) + \mathrm{tr}(A_{\mathcal{I}}B_{\mathrm{st}}(\mathrm{adj}B_{\mathrm{st}})(\mathrm{adj}A_{\mathrm{st}}))) \\
&= \det A_{\mathrm{st}}\ \det B_{\mathrm{st}} + \varepsilon(\mathrm{tr}(B_{\mathcal{I}}(\mathrm{adj}B_{\mathrm{st}})(\mathrm{adj}A_{\mathrm{st}})A_{\mathrm{st}}) + \mathrm{tr}(A_{\mathcal{I}}B_{\mathrm{st}}(\mathrm{adj}B_{\mathrm{st}})(\mathrm{adj}A_{\mathrm{st}}))) \\
&= \det A_{\mathrm{st}}\ \det B_{\mathrm{st}} + \varepsilon(\mathrm{tr}(B_{\mathcal{I}}(\mathrm{adj}B_{\mathrm{st}})(\det A_{\mathrm{st}})I_n) + \mathrm{tr}(A_{\mathcal{I}}(\det B_{\mathrm{st}})I_n(\mathrm{adj}A_{\mathrm{st}}))) \\
&= \det A_{\mathrm{st}}\ \det B_{\mathrm{st}} + \varepsilon(\det A_{\mathrm{st}}\mathrm{tr}(B_{\mathcal{I}}(\mathrm{adj}B_{\mathrm{st}})) + \det B_{\mathrm{st}}\mathrm{tr}(A_{\mathcal{I}}(\mathrm{adj}A_{\mathrm{st}}))) \\
&= \det(A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}})\det(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}) \\
&= \det\widehat{A}\ \det\widehat{B}.
\end{aligned}$$

□

Lemma 4.10. *Let $\widehat{A} := [\alpha_{ij}] + \varepsilon[\beta_{ij}] \in M_n(\mathbb{DF})$ such that $\widehat{A}$ is upper triangular. Then*

$$det\ \widehat{A} = \prod_{i=1}^{n}(\alpha_{ii} + \varepsilon\beta_{ii})$$

Proof. Since $\widehat{A}$ is upper triangular, A_{st} and $A_{\mathcal{I}}$ are upper triangular, and so $F_k(A_{\mathrm{st}}, A_{\mathcal{I}})$ is upper triangular for all $k \in \{1, 2, \ldots, n\}$. Thus,

$$\begin{aligned}
\det\widehat{A} &= \det A_{\mathrm{st}} + \varepsilon\sum_{k=1}^{n}\prod_{j\neq k}\alpha_{jj}\beta_{kk} \\
&= \prod_{i=1}^{n}\alpha_{ii} + \varepsilon\sum_{k=1}^{n}\prod_{j\neq k}\alpha_{jj}\beta_{kk} \\
&= \prod_{i=1}^{n}(\alpha_{ii} + \varepsilon\beta_{ii}).
\end{aligned}$$

□

Theorem 4.11. *Let $\widehat{A} \in M_n(\mathbb{DC})$, where A_{st} has n distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then,*

$$det\, \widehat{A} = \prod_{i=1}^{n} (\lambda_i + \varepsilon\mu_i),$$

where $\lambda_i + \varepsilon\mu_i$ is an eigenvalue of $\widehat{A}$ for all $i \in \{1, 2, \ldots, n\}$.

Proof. By Schur's triangularization theorem, we know that there exists a unitary $\widehat{U} \in M_n(\mathbb{DC})$ such that $\widehat{T} := [t_{ii}] = \widehat{U}^*\widehat{A}\widehat{U}$ and $\det \widehat{T} = \prod_{i=1}^{n} t_{ii}$, where each $t_{ii} = \lambda_i + \varepsilon\mu_i$ is an eigenvalue of $\widehat{A}$ (and of $\widehat{T}$). Thus,

$$\begin{aligned}
\det \widehat{A} &= \det\left(\widehat{U}\widehat{T}\widehat{U}^*\right) \\
&= \det\widehat{U}\ \det\widehat{T}\ \det\widehat{U}^* \\
&= \det \widehat{U} \det \widehat{U}^* \det \widehat{T} \\
&= \det(\widehat{U}\widehat{U}^*) \det \widehat{T} \\
&= \det I_n \det \widehat{T} \\
&= \det \widehat{T} \\
&= \prod_{i=1}^{n} t_{ii} \\
&= \prod_{i=1}^{n} (\lambda_i + \varepsilon\mu_i).
\end{aligned}$$

□

4.3 Sylvester's Theorem

The standard *Sylvester's theorem* on linear matrix equations provides a necessary and sufficient condition for the existence and uniqueness of the solution of the complex matrix equation $AX - XB = C$: A and B should have no eigenvalues in common [6] [7]. In this paper, we are interested on extending this theorem to dual complex matrices.

Theorem 4.12 (Sylvester's Theorem for Dual Complex Matrices)**.** *Let $\widehat{A} \in M_n(\mathbb{DC})$ and $\widehat{B} \in M_m(\mathbb{DC})$ be given. The equation $\widehat{A}\widehat{X} - \widehat{X}\widehat{B} = \widehat{C}$ has a unique solution $\widehat{X} = X_{st} + \varepsilon X_{\mathcal{I}} \in M_{n,m}(\mathbb{DC})$ for each given $\widehat{C} = C_{st} + \varepsilon C_{\mathcal{I}} \in M_{n,m}(\mathbb{DC})$ if and only if $\sigma(A_{st}) \cap \sigma(B_{st}) = \varnothing$, i.e., A_{st} and B_{st} have no eigenvalues in common.*

Proof. Consider the equation

$$\widehat{A}\widehat{X} - \widehat{X}\widehat{B} = \widehat{C} \tag{4.1}$$

which is equivalent to

$$\begin{aligned}\widehat{A}\widehat{X} - \widehat{X}\widehat{B} &= (A_{\mathrm{st}} + \varepsilon A_{\mathcal{I}})(X_{\mathrm{st}} + \varepsilon X_{\mathcal{I}}) - (X_{\mathrm{st}} + \varepsilon X_{\mathcal{I}})(B_{\mathrm{st}} + \varepsilon B_{\mathcal{I}}) \\ &= A_{\mathrm{st}}X_{\mathrm{st}} + \varepsilon(A_{\mathrm{st}}X_{\mathcal{I}} + A_{\mathcal{I}}X_{\mathrm{st}}) - (X_{\mathrm{st}}B_{\mathrm{st}} + \varepsilon(X_{\mathrm{st}}B_{\mathcal{I}} + X_{\mathcal{I}}B_{\mathrm{st}})) \\ &= A_{\mathrm{st}}X_{\mathrm{st}} - X_{\mathrm{st}}B_{\mathrm{st}} + \varepsilon(A_{\mathrm{st}}X_{\mathcal{I}} + A_{\mathcal{I}}X_{\mathrm{st}} - (X_{\mathrm{st}}B_{\mathcal{I}} + X_{\mathcal{I}}B_{\mathrm{st}})) \\ &= A_{\mathrm{st}}X_{\mathrm{st}} - X_{\mathrm{st}}B_{\mathrm{st}} + \varepsilon(A_{\mathrm{st}}X_{\mathcal{I}} - X_{\mathcal{I}}B_{\mathrm{st}} + A_{\mathcal{I}}X_{\mathrm{st}} - X_{\mathrm{st}}B_{\mathcal{I}}) \\ &= C_{\mathrm{st}} + \varepsilon C_{\mathcal{I}}\end{aligned}$$

($\rightarrow$) Assume that $\widehat{A}\widehat{X} - \widehat{X}\widehat{B} = \widehat{C}$ has a unique solution $\widehat{X} \in M_{n,m}(\mathbb{DC})$. Then from the equation above, $A_{\mathrm{st}}X_{\mathrm{st}} - X_{\mathrm{st}}B_{\mathrm{st}} = C_{\mathrm{st}}$ has a unique solution X_{st}, and so by Sylvester's theorem for matrices over $\mathbb{C}$, A_{st} and B_{st} have no eigenvalues in common.

($\leftarrow$) Suppose A_{st} and B_{st} have no eigenvalues in common. By Sylvester's theorem for matrices over $\mathbb{C}$ and from the equation above, $A_{\mathrm{st}}X_{\mathrm{st}} - X_{\mathrm{st}}B_{\mathrm{st}} = C_{\mathrm{st}}$ has a unique solution X_{st}. Note that

$$A_{\mathrm{st}}X_{\mathcal{I}} - X_{\mathcal{I}}B_{\mathrm{st}} = C_I - (A_{\mathcal{I}}X_{\mathrm{st}} - X_{\mathrm{st}}B_{\mathcal{I}}) =: D$$

and $A_{\mathrm{st}}, B_{\mathrm{st}}$, and D are given. We can apply Sylvester's theorem for matrices over $\mathbb{C}$ since A_{st} and B_{st} have no eigenvalues in common. Thus, the previous equation has a unique solution $X_{\mathcal{I}}$ and hence (4.1) has a solution $\widehat{X} = X_{\mathrm{st}} + \varepsilon X_{\mathcal{I}}$. Now, suppose that $\widehat{Y} = Y_{\mathrm{st}} + \varepsilon Y_{\mathcal{I}} \in M_n(\mathbb{DC})$ is another solution for (4.1). Then by Sylvester's theorem for matrices over $\mathbb{C}$, Y_{st} is the unique solution to $A_{\mathrm{st}}Y_{\mathrm{st}} - Y_{\mathrm{st}}B_{\mathrm{st}} = C_{\mathrm{st}}$. Hence, $X_{\mathrm{st}} = Y_{\mathrm{st}}$. Similarly, if $Y_{\mathcal{I}}$ is the unique solution to $A_{\mathrm{st}}Y_{\mathcal{I}} - Y_{\mathcal{I}}B_{\mathrm{st}} = D$, then $X_{\mathcal{I}} = Y_{\mathcal{I}}$. Since $X_{\mathrm{st}} = Y_{\mathrm{st}}$ and $X_{\mathcal{I}} = Y_{\mathcal{I}}$, we have $\widehat{X} = \widehat{Y}$. Hence $\widehat{X}$ is the unique solution to (4.1). $\square$

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